UNIVERSITY OF JAFFNA

FACULTY OF ENGINEERING



NETWORK DESIGN

EC 7091: COMMUNICATION NETWORK DESIGN

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1. INTRODUCTION

1.1 Background

Modern hospitals depend heavily on reliable communication networks to support clinical operations, administrative activities, medical information systems, real-time patient monitoring, and remote healthcare services. Unlike conventional enterprise networks, a hospital network must handle both normal data traffic and highly critical medical traffic while maintaining high availability, security, low latency, and controlled access to sensitive healthcare information.

The proposed network is designed for a hospital consisting of four main clinical areas: the ICU/Surgical Ward, Cardiology Ward, Emergency/Trauma Ward, and Outpatient Diagnostics Ward. Each ward contains wired medical workstations, wireless tablets or laptops, and telemetry or IoT-based patient-monitoring devices. In addition, the hospital includes an Administration network, a Hospital Operations Centre (HOC), and several important medical and business servers such as the Picture Archiving and Communication System (PACS), Electronic Medical Record/Electronic Health Record (EMR/EHR), Billing, Inventory, and Telemetry servers.

The network must also support high-priority services such as real-time robotic telesurgery and continuous patient telemetry streams. These services require low delay, reliable connectivity, and appropriate traffic prioritization . Furthermore, connectivity is required between the hospital, external healthcare services, the Internet, and remotely connected ambulances.

1.2 Aim of the Network Design

The aim of this project is to design and simulate a secure, scalable, redundant, and high-performance hospital network capable of supporting clinical, administrative, and real-time medical applications. The proposed network integrates departmental segmentation, redundant core infrastructure, IPv4 and IPv6 addressing, dynamic routing, secure Internet connectivity, Quality of Service (QoS), network security controls, and remote medical connectivity.

Cisco Packet Tracer is used to model and validate the proposed network architecture. The simulation is used to demonstrate logical connectivity, routing, redundancy, traffic prioritization , access restrictions, server connectivity, Internet access, and selected security mechanisms.

1.3 Scope of the Design

The scope of the proposed hospital network includes the following major components:

- Segmentation of hospital departments using Virtual Local Area Networks (VLANs).
- IPv4 subnetting and IPv6 dual-stack addressing.
- Redundant multilayer core switches for inter-VLAN routing.
- First Hop Redundancy using Hot Standby Router Protocol (HSRP).
- Dynamic WAN routing using Open Shortest Path First (OSPF).
- Primary and secondary WAN links to provide external connectivity and redundancy.

- Quality of Service policies for robotic telesurgery, patient telemetry, EMR, and PACS traffic.
- Network Address Translation/Port Address Translation (NAT/PAT) for Internet access.
- Firewall-style access control using extended Access Control Lists (ACLs).
- A Demilitarized Zone (DMZ) for externally accessible hospital services.
- A dedicated management network for the Hospital Operations Centre.
- Secure remote ambulance connectivity through a tunnel-based WAN connection.

2. NETWORK REQUIREMENTS ANALYSIS

2.1 Hospital Network Requirements

2.1.1 Clinical Departments

The hospital network consists of four major clinical departments: ICU/Surgical, Cardiology, Emergency/Trauma, and Outpatient Diagnostics. Each department requires reliable wired and wireless connectivity for clinical workstations, mobile devices, and patient-monitoring equipment. The network must provide logical separation between departments while allowing authorised communication with shared hospital services such as EMR/EHR, PACS, and telemetry systems.

Table 01 : Distribution of devices

Ward	Medical PCs	Staff Tablets/Laptops	Telemetry Nodes	Devices
ICU/Surgical	10	12	8	30
Cardiology	10	12	8	30
Emergency/Trauma	10	12	8	30
Outpatient Diagnostics	10	12	8	30
Total	40	48	32	120

2.1.2 Administration Network

The Administration department contains five PCs and requires access to hospital business systems, particularly the Billing and Inventory servers. For security purposes, administrative users are logically separated from clinical departments. Direct Internet access from the Administration network is restricted according to the hospital security policy.

2.1.3 Hospital Operations Centre

The Hospital Operations Centre (HOC) is responsible for monitoring and managing the hospital's network infrastructure. The HOC requires controlled access to network devices, management interfaces, and operational systems. A dedicated management network is therefore provided to separate infrastructure management traffic from normal user traffic.

2.1.4 Medical Servers

Table 02 : Medical Servers Function And Allocated Network

Server	Main Function	Network
PACS	Medical image storage and retrieval	Server VLAN
EMR/EHR	Electronic patient records	Server VLAN
Billing	Financial and billing operations	Server VLAN
Inventory	Medical equipment and inventory management	Server VLAN
Telemetry	Patient monitoring and telemetry data	Server VLAN

2.2 Real-Time Medical Traffic Requirements

2.2.1 Robotic Telesurgery

The robotic telesurgery system represents a highly time-sensitive application. Communication between the telesurgery equipment and the required remote systems must be provided with low delay, low jitter, sufficient bandwidth, and high availability. Packet loss and excessive congestion can negatively affect the responsiveness of real-time communication. Therefore, the network design must provide traffic prioritisation and redundant connectivity for telesurgery traffic.

2.2.2 Patient Telemetry

Patient telemetry devices continuously transmit monitoring information to the hospital telemetry system. The network must therefore provide reliable and continuous connectivity for telemetry traffic. Telemetry traffic is separated from ordinary user traffic through VLAN segmentation and is assigned a dedicated QoS class to reduce the effects of congestion.

2.3 Network Availability Requirements

High availability is essential because interruption of hospital network services can affect clinical operations and access to medical information. The proposed network therefore incorporates redundancy at multiple levels. Two multilayer core switches are used to reduce dependence on a single core device, while HSRP provides redundant default gateways. Redundant WAN connectivity is also provided through a primary dedicated fibre link and a secondary microwave link. Dynamic routing allows traffic to use the available path when a primary connection becomes unavailable.

2.4 Security Requirements

The hospital network handles sensitive clinical and operational information and therefore requires strong network segmentation and access control. Clinical, administrative, management, server, and public-facing services should not share the same unrestricted network. The design therefore uses VLAN segmentation, firewall ACLs, NAT/PAT, a dedicated management network, and a DMZ.

The DMZ provides an isolated network for public-facing services. External users can access authorised services such as the hospital web server, while DMZ systems are prevented from initiating unauthorised connections toward sensitive internal systems such as EMR and PACS.

2.5 Remote Ambulance Requirements

The network must support communication between a remote ambulance and selected hospital services. The ambulance should be able to access essential systems such as EMR and patient telemetry while access to sensitive infrastructure and unrelated services should remain restricted. A tunnel-based connection is therefore used to provide controlled remote connectivity between the ambulance network and the hospital network.

2.6 Internet and WAN Requirements

The hospital requires reliable Internet and external connectivity for approved services. A primary dedicated fibre connection is selected to provide high bandwidth and low latency, while a secondary licensed microwave connection provides an independent backup path. OSPF is used to support dynamic route selection and failover between the available WAN paths.

2.7 IPv4 and IPv6 Requirements

The network is designed to support both IPv4 and IPv6. IPv4 subnetting is used to allocate appropriately sized networks to the different hospital departments, while IPv6 `/64` networks are assigned to individual VLANs using the allocated `2001:DB8:ABCD::/48` address block. The dual-stack approach allows existing IPv4 services to continue operating while providing a structured IPv6 addressing scheme for future expansion.

2.8 Scalability Requirements

The network should be capable of supporting future expansion of clinical devices, IoT equipment, users, servers, and network services. VLAN-based segmentation, hierarchical addressing, redundant core infrastructure, and IPv6 addressing provide a foundation for future growth. The proposed subnet sizes also provide additional host capacity within the allocated address space.

3. PROPOSED NETWORK ARCHITECTURE AND IP NETWORK DESIGN DIAGRAM

3.1 Overall Network Architecture

The proposed hospital network uses a hierarchical and segmented architecture consisting of access networks, redundant multilayer core switches, a firewall/gateway, WAN connectivity, a DMZ, servers, and a remote ambulance connection. The network is divided into separate VLANs for the clinical departments, Administration, Hospital Operations Centre (HOC), servers, DMZ, and network management. This segmentation improves security, reduces broadcast traffic, and allows access policies to be applied according to the role of each network.

The two multilayer core switches provide inter-VLAN routing and gateway redundancy using HSRP. The hospital also uses two WAN paths: a primary dedicated fibre connection and a secondary microwave connection. OSPF is used for dynamic routing and allows the secondary path to provide connectivity if the primary path becomes unavailable.

3.2 Network Architecture Components

Table 03 : Network Architecture Components and purpose of use

Component	Purpose
Access switches	Connect clinical workstations, wireless devices and IoT/telemetry devices
CORE-SW1	Primary multilayer core switch and Layer 3 routing
CORE-SW2	Redundant multilayer core switch
HSRP	Provides redundant default gateways
FW-GATEWAY	Firewall functions, ACL enforcement and NAT/PAT
HOSPITAL-EDGE	Provides connectivity toward the WAN/Internet
Primary fibre link	Main 2 Gbps WAN connection
Secondary microwave link	Backup 1 Gbps WAN connection
Server network	Hosts PACS, EMR, Billing, Inventory and Telemetry servers
DMZ	Isolates the externally accessible hospital web service
AMBULANCE-RTR	Provides secure remote connectivity for ambulance services
HOC	Provides network and infrastructure management

3.3 IP Network Design Diagram

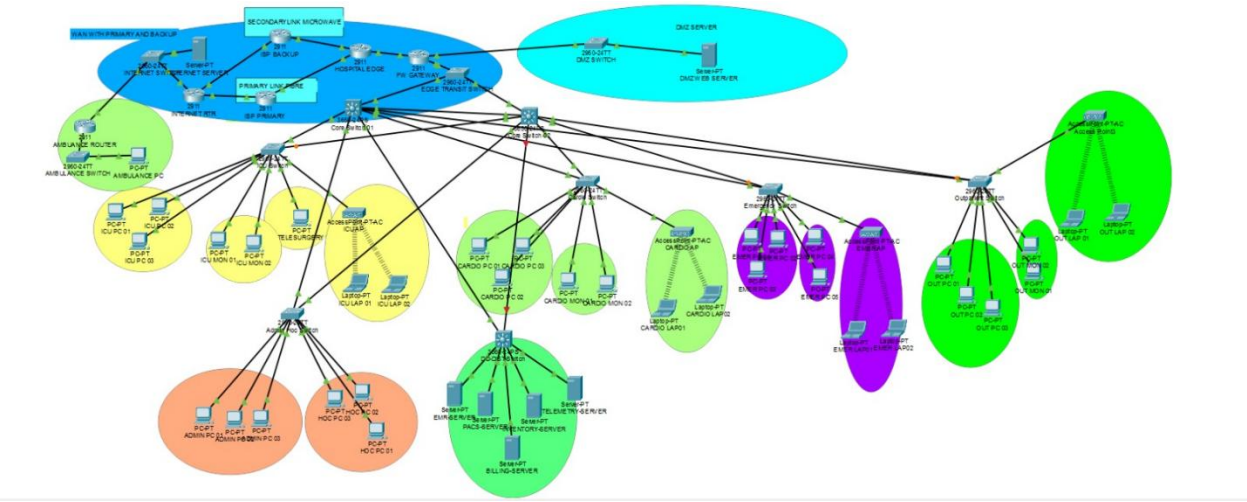


Figure 01 :Full Network Design Diagram Of Hospital And WAN

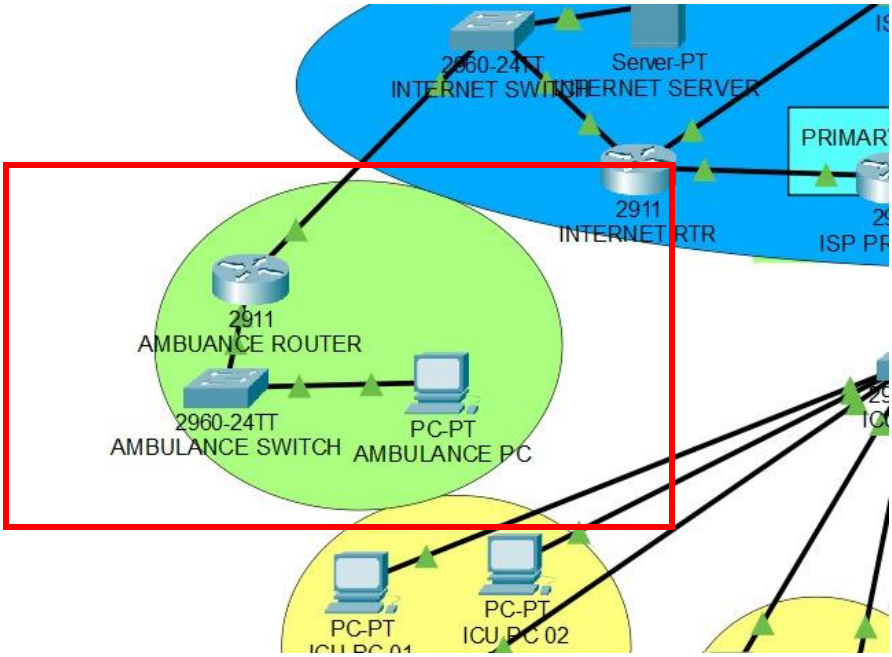


Figure 02 : Ambulance Command unit

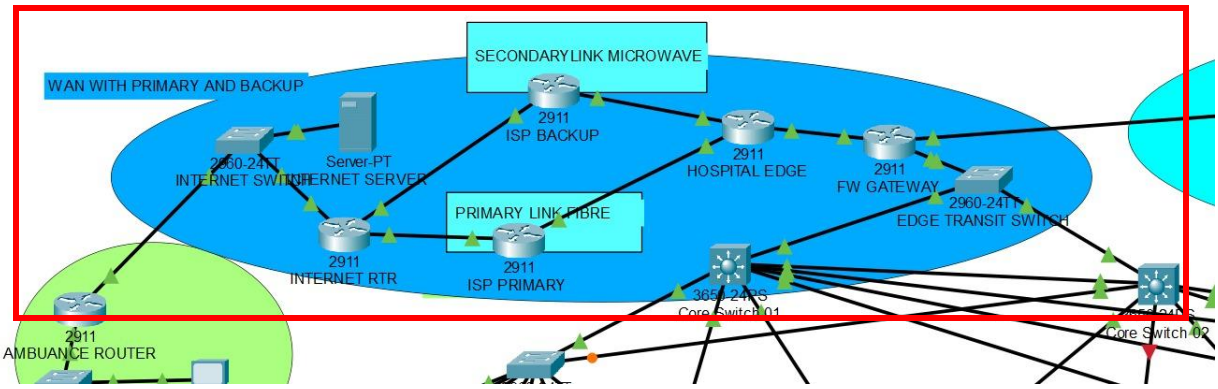


Figure 03 : WAN With Primary Access Link

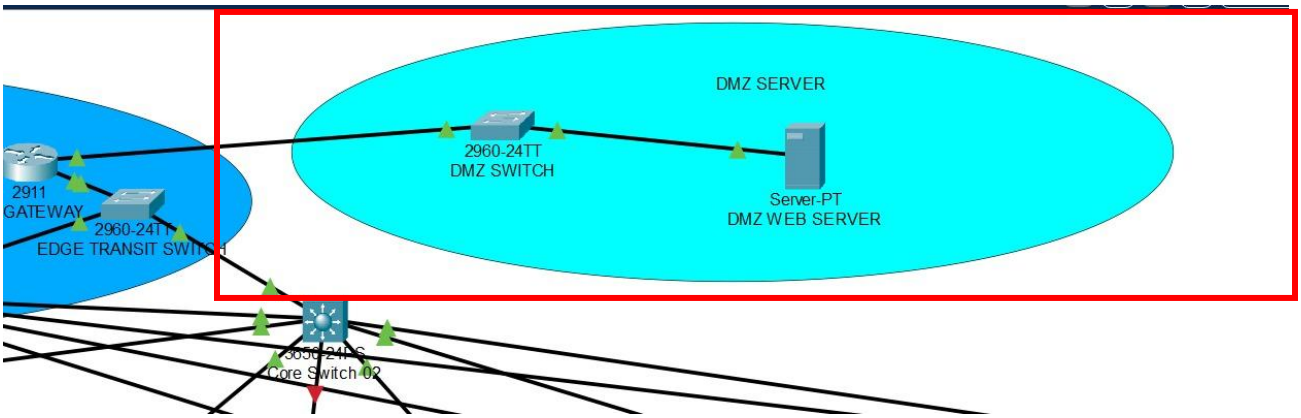


Figure 04 : DMZ Web Server

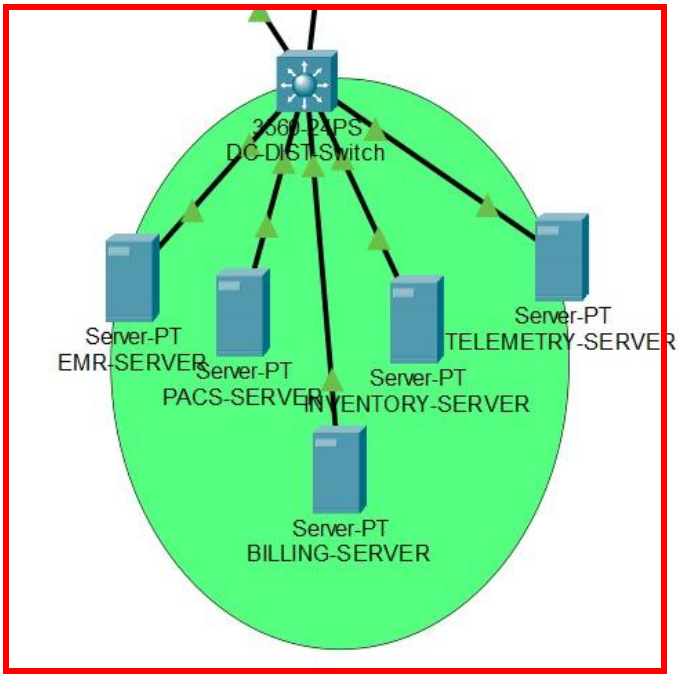


Figure 05 : Server-PT Billing Server

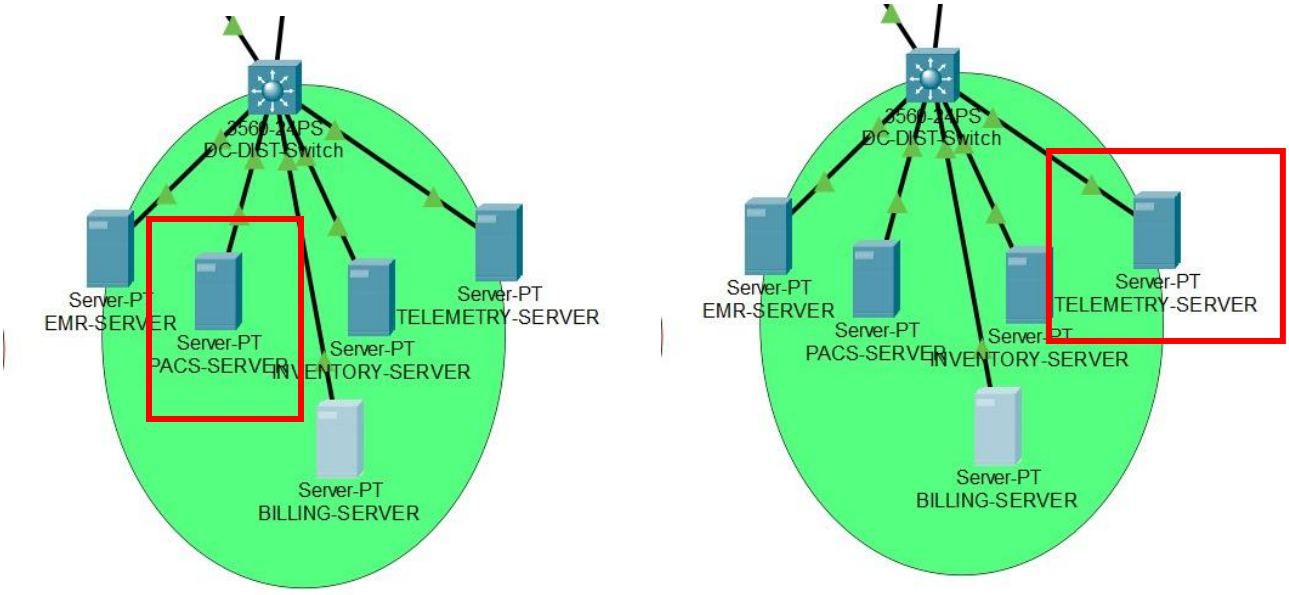


Figure 06 : DICOM Medical Imaging And Real-Time Patient Telemetry

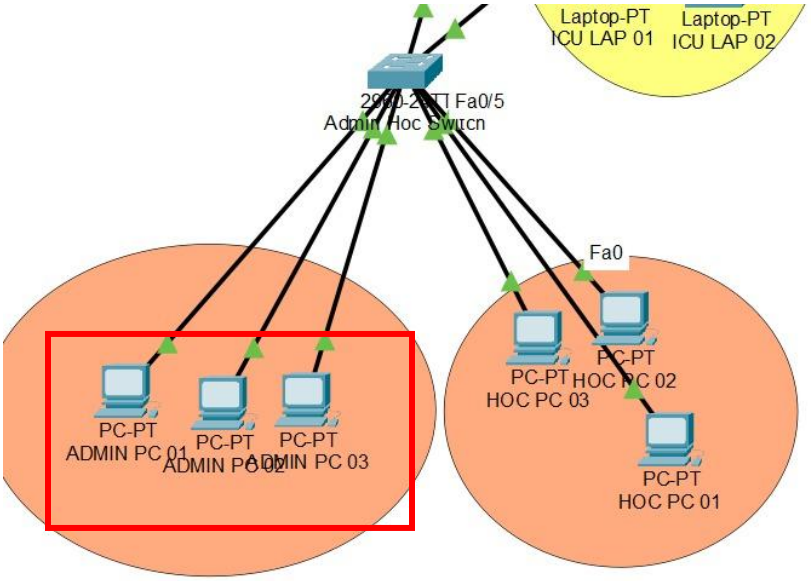


Figure 07 : Administrative Systems

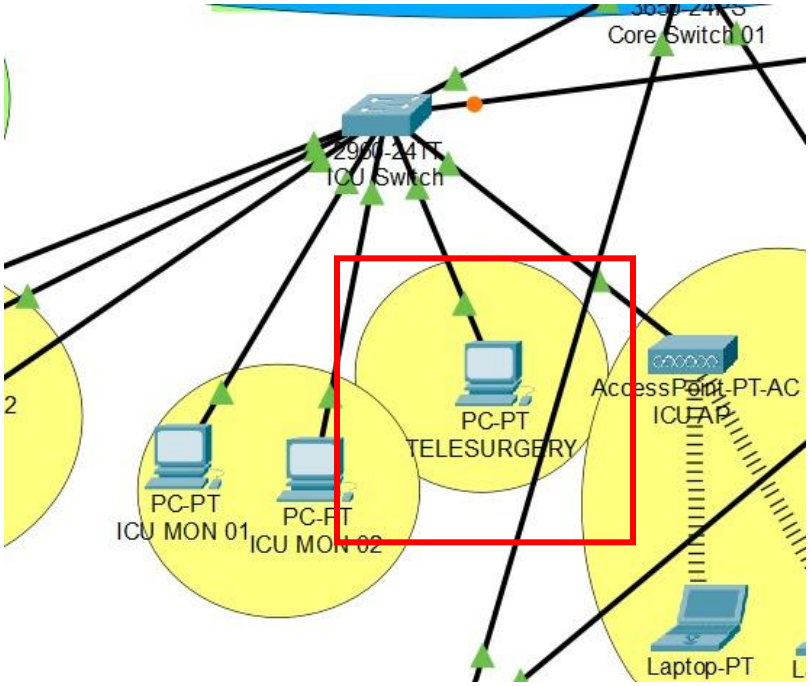


Figure 08 : Remote Telesurgery Feeds

3.4 IP Addressing and Gateway Design

Table 04: IP Addressing,Gateway , Usable Range And Broadcast

VLAN	Network	Subnet	Mask	Gateway	Usable Range	Broadcast
10	ICU/Surgical	198.18.0.0/26	255.255.255.192	198.18.0.1	.1 – .62	198.18.0.63
20	Cardiology	198.18.0.64/26	255.255.255.192	198.18.0.65	.65 – .126	198.18.0.127
30	Emergency/Trauma	198.18.0.128/26	255.255.255.192	198.18.0.129	.129 – .190	198.18.0.191
40	Outpatient	198.18.0.192/26	255.255.255.192	198.18.0.193	.193 – .254	198.18.0.255
50	Administration	198.18.1.0/28	255.255.255.240	198.18.1.1	.1 – .14	198.18.1.15
60	HOC	198.18.1.16/28	255.255.255.240	198.18.1.17	.17 – .30	198.18.1.31
70	Servers	198.18.1.32/27	255.255.255.224	198.18.1.33	.33 – .62	198.18.1.63
80	DMZ	198.18.1.64/28	255.255.255.240	198.18.1.65	.65 – .78	198.18.1.79
90	Management	198.18.1.80/28	255.255.255.240	198.18.1.81	.81 – .94	198.18.1.95
100	Ambulance/VPN	198.18.1.96/28	255.255.255.240	198.18.1.97	.97 – .110	198.18.1.111

3.5 Server and Special-Purpose IP Addresses

Table 05 : Special-Purpose IP Addresses And Functions

Device	IP Address	Function
PACS Server	198.18.1.40	Medical imaging/storage
EMR Server	198.18.1.41	Electronic medical records
Billing Server	198.18.1.42	Billing services
Inventory Server	198.18.1.43	Inventory management
Telemetry Server	198.18.1.44	Patient telemetry
DMZ Web Server	198.18.1.66	External web service
Telesurgery Terminal	198.18.0.30	Robotic telesurgery traffic

3.6 IPv6 Network Design

The network also supports dual-stack IPv4/IPv6 operation. The hospital has been allocated the IPv6 prefix: 2001:DB8:ABCD::/48

Table 06 : Individual /64 networks are assigned to the VLANs

VLAN	IPv6 Network
VLAN 10	2001:DB8:ABCD:10::/64
VLAN 20	2001:DB8:ABCD:20::/64
VLAN 30	2001:DB8:ABCD:30::/64
VLAN 40	2001:DB8:ABCD:40::/64
VLAN 50	2001:DB8:ABCD:50::/64
VLAN 60	2001:DB8:ABCD:60::/64
VLAN 70	2001:DB8:ABCD:70::/64
VLAN 80	2001:DB8:ABCD:80::/64
VLAN 90	2001:DB8:ABCD:90::/64

3.7 Regional Health Database Connectivity

The proposed network must provide reliable external connectivity to regional healthcare databases and other authorized medical services. This requirement is supported through the hospital edge infrastructure, where external traffic passes through the firewall gateway and the redundant WAN architecture. The primary fibre connection provides the preferred path, while the secondary microwave connection provides continuity during a primary-link failure. Access to external healthcare services is controlled through routing, firewall policies, and network address translation.

4. ACCESS NETWORK TECHNOLOGY COMPARISON AND SELECTION

4.1 Access Network Requirements

The hospital requires an access network capable of supporting high-bandwidth and delay-sensitive healthcare applications, including DICOM medical imaging, access to regional healthcare databases, real-time patient telemetry, and remote telesurgery communications. The selected access technology must therefore provide high bandwidth, low latency, high availability, scalability, and sufficient reliability for continuous hospital operation.

Because failure of a single external link could interrupt critical healthcare services, both a primary access link and an independent secondary access link are required. The secondary link should preferably use a different physical transmission medium so that a fibre cable failure does not simultaneously affect both connections.

4.2 Comparison of Access Network Technologies

Table 07 : Comparison of Access Network Technologies

Technology	Advantages	Limitations	Suitability for Hospital
Dedicated Fibre / DIA	Very high bandwidth; low latency and jitter; symmetric capacity; highly scalable; suitable for continuous high-volume traffic	Higher installation and subscription cost; physical fibre cuts can cause outages	Excellent – selected as primary access link
Point-to-Point Microwave	High bandwidth; low latency; independent of underground fibre infrastructure; rapid	Requires line-of-sight; weather and interference must be considered; lower capacity than fibre	Very good – selected as secondary access link

	deployment; useful for physical path diversity		
4G/5G Cellular	Rapid deployment; mobility support; wide-area coverage; useful for vehicles and emergency connectivity	Performance depends on signal strength, congestion and provider coverage; bandwidth and latency can vary	Suitable for mobile/tertiary connectivity
Satellite / LEO Satellite	Available in remote locations where terrestrial networks are unavailable; useful during disasters and for remote ambulances	Higher or variable latency; weather/visibility considerations; service cost; not ideal as the main hospital connection	Suitable as remote/backup connectivity
Shared Fibre/PON Broadband	High downstream capacity; comparatively economical; widely available	Bandwidth may be shared among users; SLA and performance guarantees can be weaker than dedicated fibre	Suitable for non-critical/general access, but not preferred for primary clinical connectivity

4.3 Selected Primary Access Link – Dedicated Fibre

- A 2 Gbps symmetric dedicated fibre connection was selected as the hospital's primary access link.

  Dedicated fibre is appropriate because the hospital generates a mixture of normal enterprise traffic and bandwidth-intensive clinical traffic. Large PACS/DICOM files generated by MRI, CT and X-ray systems can require significant network capacity, while applications such as robotic telesurgery and real-time healthcare communication are sensitive to latency and jitter.

The primary fibre connection provides:

- High symmetric bandwidth for both upload and download traffic.
- Low latency and low jitter.
- Capacity for high-volume PACS and medical data transfers.
- Support for real-time telesurgery and other interactive clinical applications.
- Scalability for future increases in medical devices and digital healthcare services.
- Better service-level guarantees than shared broadband connections.
-

Therefore, dedicated fibre provides the most appropriate primary connection for the hospital's critical external communications.

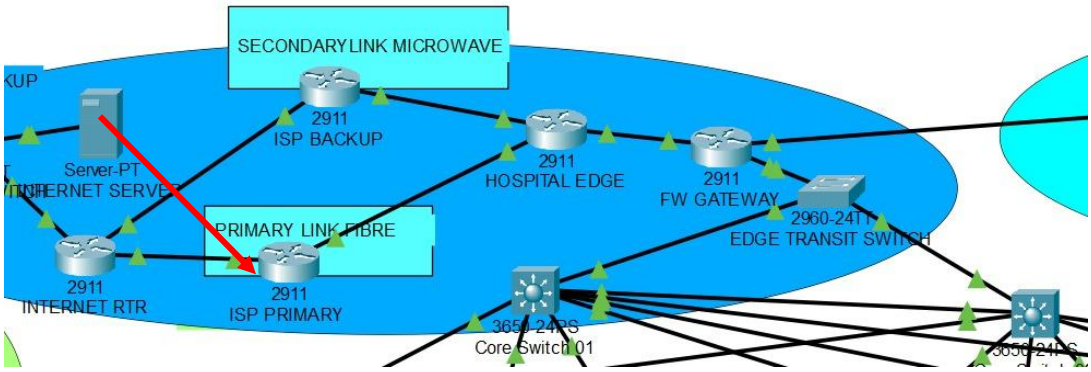


Figure 03 : Primary Dedicated Fibre Access Link Implemented in Cisco Packet Tracer

4.4 Selected Secondary Access Link – Point-to-Point Microwave

- A 1 Gbps licensed point-to-point microwave connection was selected as the secondary hospital access link.

Using another fibre service as the backup could still leave the hospital vulnerable if both fibre services share the same physical duct or cable route. Microwave provides physical-medium diversity because it does not rely on the underground fibre path.

The secondary microwave connection provides:

- An independent communication medium from the primary fibre link.
- High enough capacity to maintain important services during fibre failure.
- Relatively low latency compared with traditional satellite connectivity.
- Rapid restoration of external connectivity through automatic routing failover.
- Greater resilience against fibre cuts or civil-construction damage.

The microwave service therefore provides an appropriate disaster-resilient backup path.

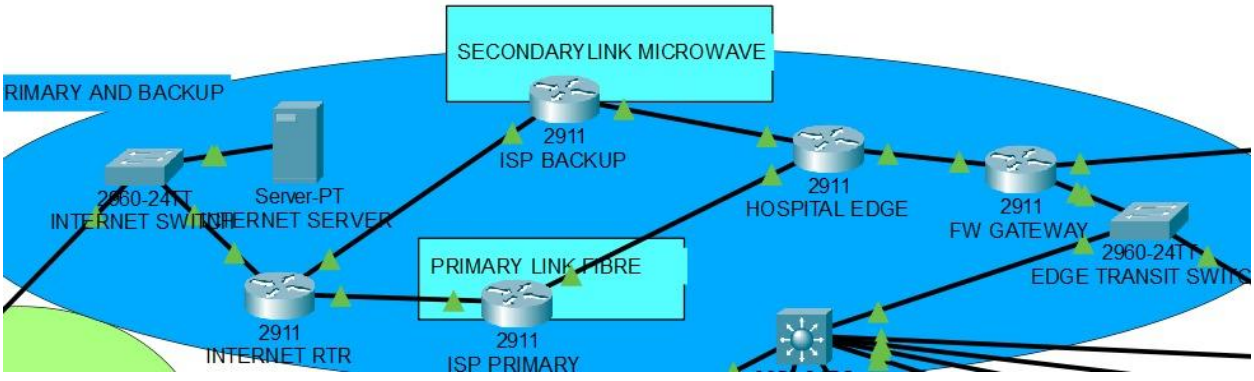


Figure 4.: Secondary Microwave Access Link Implemented in Cisco Packet Tracer

4.5 Primary and Secondary Link Redundancy

The two access technologies were combined to provide a redundant external connectivity architecture:

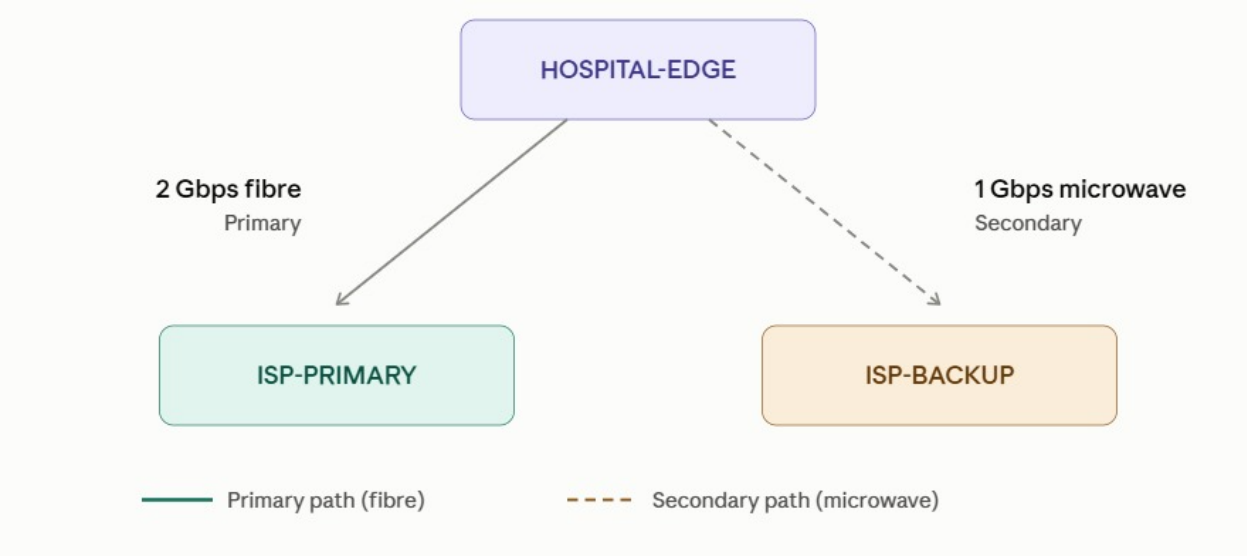


Figure 05 : Primary And Secondary Link Architecture

Under normal operating conditions, traffic uses the primary fibre link. The primary route was assigned a lower OSPF cost so that it is preferred over the secondary route. The microwave connection is assigned a higher OSPF cost and therefore remains available as a backup path. In the implemented network:

Primary Fibre

OSPF Cost = 10

Secondary Microwave

OSPF Cost = 100

If the primary connection becomes unavailable, OSPF recalculates the preferred route and traffic is forwarded through the secondary connection.

This architecture improves the availability of services such as:

- EMR/EHR connectivity
- PACS-related external communication
- Patient telemetry
- Robotic telesurgery communications
- Regional healthcare database access

4.6 Implementation and Failover Verification

The primary and secondary WAN connections were implemented in Cisco Packet Tracer using separate interfaces on HOSPITAL-EDGE. OSPF was configured to prefer the fibre path during normal operation while retaining the microwave link as an alternate route. For evidence, include the following screenshots.

```
Router>
Router>enable
Router#show ip route 192.0.2.128
Routing entry for 192.0.2.128/25
Known via "ospf 1", distance 110, metric 12, type intra area
  Last update from 203.0.113.2 on GigabitEthernet0/1, 02:20:14 ago
  Routing Descriptor Blocks:
  * 203.0.113.2, from 192.0.2.129, 02:20:14 ago, via GigabitEthernet0/1
      Route metric is 12, traffic share count is 1

Router#
```

Figure 6: Preferred Route Through the Primary Fibre Connection

```
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gi0/1
Router(config-if)#shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

05:39:17: %OSPF-5-ADJCHG: Process 1, Nbr 203.0.113.2 on GigabitEthernet0/1 from FULL to DOWN,
Neighbor Down: Interface down or detached

Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Figure 07 : Shutdown The Fibre Connection To Test Whether The Backup Connection Work Or Not

```
Router#show ip route 192.0.2.128
Routing entry for 192.0.2.128/25
Known via "ospf 1", distance 110, metric 102, type intra area
  Last update from 198.51.100.2 on GigabitEthernet0/2, 00:00:26 ago
  Routing Descriptor Blocks:
  * 198.51.100.2, from 192.0.2.129, 00:00:26 ago, via GigabitEthernet0/2
      Route metric is 102, traffic share count is 1

Router#
```

Figure 08 : Testing The Backup Connection

```
Router#ping 192.0.2.130

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.0.2.130, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

Router#
```

Figure 09 : From The Hospital Edge Send The ICMP Packets To The Internetserver

- now the packets are using the backup link because we shutdown the primary link

4.7 Mobile and Emergency Access Technologies

Although fibre and microwave were selected for the fixed hospital campus, mobile and emergency environments have different requirements.

4G/5G cellular connectivity is appropriate for the Remote Mobile Ambulance because the ambulance continuously changes location and therefore cannot depend on fixed wired infrastructure. Cellular connectivity provides the primary mobile WAN connection for the ambulance.

Satellite connectivity can provide an additional fallback where terrestrial cellular coverage is unavailable. It is particularly useful in remote locations or during large-scale disasters. However, due to variable latency and available capacity, satellite connectivity should not automatically be assumed suitable for delay-critical robotic control without application-specific testing.

In the Packet Tracer implementation, the remote ambulance network is represented through external WAN connectivity and a GRE tunnel to the hospital network.

4.8 Access Network Selection Summary

Table 08 : Access Network Selection Summary

Purpose	Selected Technology	Reason
Hospital primary WAN	2 Gbps Dedicated Fibre	High capacity, low latency, reliability and scalability
Hospital secondary WAN	1 Gbps Point-to-Point Microwave	Independent physical medium and backup connectivity
Mobile ambulance	4G/5G Cellular	Mobility and broad-area coverage
Remote/disaster fallback	Satellite/LEO	Connectivity where terrestrial infrastructure is unavailable
Ward wireless access	Wi-Fi 6	Secure high-speed wireless connectivity for clinical staff

5. CORE NETWORK TECHNOLOGY COMPARISON AND SELECTION

5.1 Core Network Requirements

The hospital core network must provide fast, reliable, and secure communication between the four clinical wards, Administration network, Hospital Operations Centre (HOC), centralized Data Centre, DMZ, and external WAN infrastructure. Since the network carries both conventional enterprise traffic and critical healthcare traffic, the core must support high throughput, low latency, network segmentation, redundancy, dynamic routing, and Quality of Service.

The core network must also support services such as PACS medical imaging, EMR/EHR access, continuous patient telemetry, and remote telesurgery. Therefore, the selected architecture should provide sufficient performance while avoiding unnecessary complexity and cost.

5.2 Comparison of Core Network Technologies

Table 09 – Comparison of Core Network Technologies

Core Technology	Advantages	Limitations	Suitability for Hospital
Layer 2 Ethernet with STP	Simple implementation; low equipment cost; supports VLANs	Large Layer 2 domains; blocked redundant links; slower recovery; limited scalability	Suitable only for access-layer networks
Layer 3 Switched Ethernet	High-speed hardware routing; strong VLAN segmentation; supports OSPF, ACLs, HSRP and QoS; scalable	Requires routing and redundancy configuration	Excellent -selected for hospital core
MPLS Core	Strong traffic engineering; VPN support; scalable for geographically distributed sites	Greater cost and operational complexity; normally provider-oriented	More suitable for large multi-site healthcare networks
EVPN/VXLAN Fabric	Highly scalable; flexible segmentation; supports modern data-centre fabrics	High implementation and management complexity; requires advanced hardware and expertise	Suitable for much larger hospital/data-centre environments
Traditional Router-Based Core	Clear Layer 3 boundaries; mature routing features	Lower port density and potentially higher cost compared with multilayer switching	Technically suitable but inefficient for this campus size

5.3 Selected Core Technology – Layer 3 Switched Ethernet

- A Layer 3 switched Ethernet architecture was selected as the hospital core network.

Multilayer switches combine high-speed Ethernet switching with Layer 3 routing functionality. This makes them suitable for a hospital campus where multiple VLANs must communicate through a centralized routing infrastructure.

The selected core provides:

- High-speed packet forwarding.
- Inter-VLAN routing.
- IPv4 and IPv6 support.
- Dynamic routing using OSPF.
- Gateway redundancy using HSRP.
- Access Control List enforcement.
- Quality of Service support.
- VLAN and trunk management.
- Scalability for additional wards, servers, and medical devices.

Compared with MPLS or EVPN/VXLAN, Layer 3 switched Ethernet provides the required hospital functionality with significantly less operational complexity. Therefore, it provides an appropriate balance between performance, availability, scalability, security, and cost.

5.4 Redundant Core Architecture

Two multilayer switches were deployed:

- CORE-SW1
- CORE-SW2

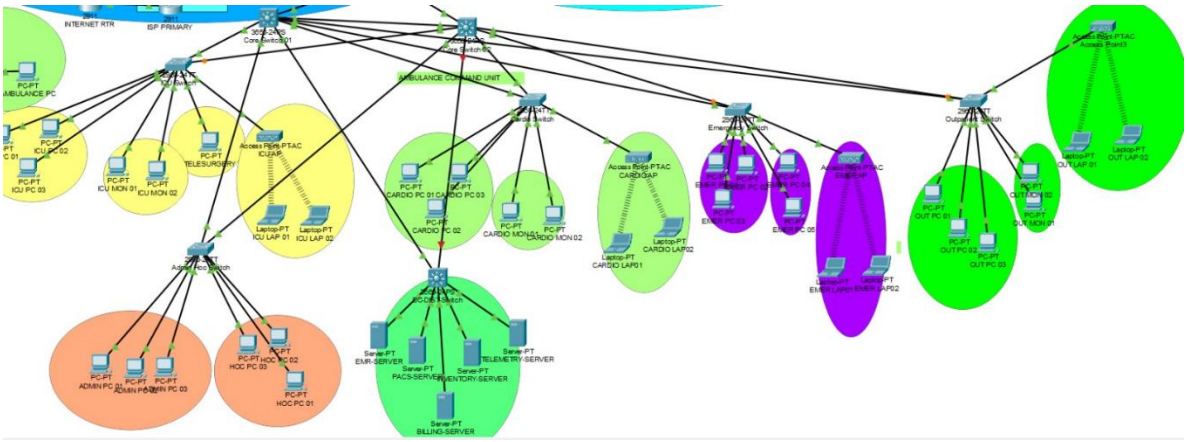


Figure 10 : Redundant Layer 3 Core Architecture Implemented in Cisco Packet Tracer

5.5 First-Hop Redundancy Using HSRP

Hot Standby Router Protocol (HSRP) was implemented to provide redundant default gateways for the hospital VLANs.
Instead of configuring client devices to use the physical IP address of either CORE-SW1 or CORE-SW2, each VLAN uses a virtual gateway address.

- For example, ICU VLAN 10 uses:
  Virtual Gateway : 198.18.0.1
  CORE-SW1 : 198.18.0.2
  CORE-SW2 : 198.18.0.3
- Under normal operation:
  CORE-SW1 → Active
  CORE-SW2 → Standby

If the active core becomes unavailable, the standby core can assume the virtual gateway role. This improves network availability because end devices do not need to change their configured gateway following a core-switch failure.

```
Switch>enable
Switch#show standby brief
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State    Active          Standby         Virtual IP
Vl10        10   110 P Active   local           198.18.0.3      198.18.0.1
Vl20        20   110 P Active   local           198.18.0.67     198.18.0.65
Vl30        30   110 P Active   local           198.18.0.131    198.18.0.129
Vl40        40   110 P Active   local           198.18.0.195    198.18.0.193
Vl50        50   110 P Active   local           198.18.1.3      198.18.1.1
Vl60        60   110 P Active   local           198.18.1.19     198.18.1.17
Vl70        70   110 P Active   local           198.18.1.35     198.18.1.33
Vl90        90   110 P Active   local           198.18.1.83     198.18.1.81
Vl999       99   110 P Active   local           10.255.0.3      10.255.0.1
Switch#
```

Figure 11 : HSRP Gateway Redundancy On The Hospital Core Network

5.6 VLAN and Inter-VLAN Routing Support

Table 10 : VLAN And Inter-VLAN Routing Support

VLAN	Function
VLAN 10	ICU/Surgical
VLAN 20	Cardiology
VLAN 30	Emergency/Trauma
VLAN 40	Outpatient Diagnostics
VLAN 50	Administration
VLAN 60	HOC
VLAN 70	Centralized Data Centre
VLAN 90	Network Management

```
Switch#
Switch#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet1/0/1   unassigned      YES unset  up                    up
GigabitEthernet1/0/2   unassigned      YES unset  up                    up
GigabitEthernet1/0/3   unassigned      YES unset  up                    up
GigabitEthernet1/0/4   unassigned      YES unset  up                    up
GigabitEthernet1/0/5   unassigned      YES unset  up                    up
GigabitEthernet1/0/6   unassigned      YES unset  up                    up
GigabitEthernet1/0/7   unassigned      YES unset  up                    up
GigabitEthernet1/0/8   unassigned      YES unset  up                    up
GigabitEthernet1/0/9   unassigned      YES unset  down                  down
GigabitEthernet1/0/10  unassigned      YES unset  down                  down
GigabitEthernet1/0/11  unassigned      YES unset  down                  down
GigabitEthernet1/0/12  unassigned      YES unset  down                  down
GigabitEthernet1/0/13  unassigned      YES unset  down                  down
GigabitEthernet1/0/14  unassigned      YES unset  down                  down
GigabitEthernet1/0/15  unassigned      YES unset  down                  down
GigabitEthernet1/0/16  unassigned      YES unset  down                  down
GigabitEthernet1/0/17  unassigned      YES unset  down                  down
GigabitEthernet1/0/18  unassigned      YES unset  down                  down
GigabitEthernet1/0/19  unassigned      YES unset  down                  down
GigabitEthernet1/0/20  unassigned      YES unset  down                  down
GigabitEthernet1/0/21  unassigned      YES unset  down                  down
GigabitEthernet1/0/22  unassigned      YES unset  down                  down
GigabitEthernet1/0/23  unassigned      YES unset  down                  down
GigabitEthernet1/0/24  unassigned      YES unset  down                  down
GigabitEthernet1/1/1   unassigned      YES unset  down                  down
GigabitEthernet1/1/2   unassigned      YES unset  down                  down
GigabitEthernet1/1/3   unassigned      YES unset  down                  down
GigabitEthernet1/1/4   unassigned      YES unset  down                  down
Vlan1                  unassigned      YES unset  administratively down down
Vlan10                 198.18.0.2      YES manual up                    up
Vlan20                 198.18.0.66     YES manual up                    up
Vlan30                 198.18.0.130    YES manual up                    up
Vlan40                 198.18.0.194    YES manual up                    up
Vlan50                 198.18.1.2      YES manual up                    up
Vlan60                 198.18.1.18     YES manual up                    up
Vlan70                 198.18.1.34     YES manual up                    up
Vlan90                 198.18.1.82     YES manual up                    up
Vlan999                10.255.0.2      YES manual up                    up
Switch#
Switch#
```

Figure 12 : Layer 3 VLAN Interfaces Configured on CORE-SW1

```
Switch#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig1/0/1    on           802.1q         trunking      1
Gig1/0/2    on           802.1q         trunking      1
Gig1/0/3    on           802.1q         trunking      1
Gig1/0/4    on           802.1q         trunking      1
Gig1/0/5    on           802.1q         trunking      1
Gig1/0/6    on           802.1q         trunking      1
Gig1/0/7    on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gig1/0/1    10,90
Gig1/0/2    10,20,30,40,50,60,70,90
Gig1/0/3    20,90
Gig1/0/4    30,90
Gig1/0/5    40,90
Gig1/0/6    50,60,90
Gig1/0/7    70,90

Port        Vlans allowed and active in management domain
Gig1/0/1    10,90
Gig1/0/2    10,20,30,40,50,60,70,90
Gig1/0/3    20,90
Gig1/0/4    30,90
Gig1/0/5    40,90
Gig1/0/6    50,60,90
Gig1/0/7    70,90

Port        Vlans in spanning tree forwarding state and not pruned
Gig1/0/1    10,90
Gig1/0/2    10,20,30,40,50,60,70,90
Gig1/0/3    20,90
Gig1/0/4    30,90
Gig1/0/5    40,90
Gig1/0/6    50,60,90
Gig1/0/7    70,90

Switch#
Switch#
```

Figure 13 : VLAN Trunk Connectivity Between the Core and Access Networks

5.7 Centralized Data Centre Integration

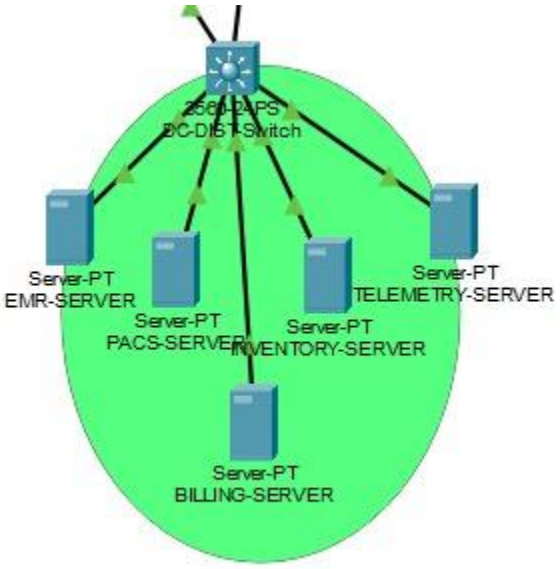


Figure 14 : Centralized Hospital Data Centre Connected to the Core Network

5.8 Support for Critical Medical Traffic

- One of the major reasons for selecting a Layer 3 switched core is its ability to support QoS and route critical medical traffic efficiently.
- The network carries different traffic types with different performance requirements.

Table 11: application and performance requirements

Application	Main Requirement
Robotic telesurgery	Very low delay and jitter
Patient telemetry	Continuous and reliable delivery
PACS/DICOM	High bandwidth
EMR/EHR	Reliable and secure access
Administration	Normal enterprise connectivity

The network therefore uses QoS to differentiate these traffic classes.

In the implemented design:

- Telesurgery → Strict Priority / 30%
- Telemetry → 10%
- EMR → 15%
- PACS → 20%

This enables delay-sensitive traffic to receive preferential treatment during periods of congestion. Detailed QoS configuration and testing can be discussed later in the dedicated QoS section, so here you only need to explain why the core infrastructure is capable of supporting it.

5.9 Spanning Tree and Layer 2 Loop Prevention

- Where redundant Layer 2 links exist between switches, Spanning Tree Protocol is used to prevent switching loops.
- STP ensures that only appropriate Layer 2 paths remain forwarding while redundant paths can be maintained without creating broadcast loops.
- During the Packet Tracer implementation, the redundant switching infrastructure was configured so that VLAN forwarding remained stable while preventing Layer 2 looping.

```
Switch#
Switch#show spanning-tree vlan 90
VLAN0090
  Spanning tree enabled protocol ieee
  Root ID    Priority    24666
             Address     0006.2AD5.E4E1
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    24666  (priority 24576 sys-id-ext 90)
             Address     0006.2AD5.E4E1
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Gi1/0/3          Desg FWD 19        128.3    P2p
Gi1/0/4          Desg FWD 19        128.4    P2p
Gi1/0/2          Desg FWD 4         128.2    P2p
Gi1/0/5          Desg FWD 19        128.5    P2p
Gi1/0/6          Desg FWD 19        128.6    P2p
Gi1/0/7          Desg FWD 19        128.7    P2p
Gi1/0/1          Desg FWD 19        128.1    P2p

Switch#
```

Figure 15 : spanning tree configuration

6. DATA TRANSMISSION TECHNOLOGY COMPARISON AND SELECTION

6.1 Transmission Requirements

The hospital network uses different transmission technologies because each part of the network has different requirements. Clinical workstations and patient monitors require reliable wired connectivity, staff tablets require wireless mobility, the campus backbone requires high-capacity links, and external/remote connections require resilient WAN technologies.

6.2 Data Transmission Technology Comparison

Table 12 : Data Transmission Technology Comparison

Technology	Main Advantages	Main Limitations	Selected Use
Copper Ethernet (Cat6/Cat6A)	Low cost, reliable, easy to deploy, supports 1 Gb/s and PoE	Limited distance compared with fibre	PCs, monitors, AP connections
Single-Mode Fibre	Very high bandwidth, long distance, low interference, scalable	Higher installation cost	Campus backbone and primary WAN

Wi-Fi 6	Mobility, high wireless capacity, suitable for tablets/laptops	Shared medium, interference and coverage dependence	Clinical staff wireless access
Microwave Radio	Independent from fibre path, low latency, rapid deployment	Requires line-of-sight and careful link planning	Secondary WAN
4G/5G Cellular	Mobile connectivity, wide coverage	Variable performance and congestion	Ambulance connectivity
Satellite	Works where terrestrial networks are unavailable	Higher/variable latency and cost	Ambulance/disaster fallback

6.3 Selected Transmission Technologies

For the hospital LAN, Ethernet over copper is used for medical workstations, patient-monitoring devices, and access-point connections because it provides reliable and cost-effective local connectivity.

For the campus backbone and high-bandwidth server communication, optical fibre is preferred because it provides significantly greater capacity, lower susceptibility to electromagnetic interference, and better scalability. Wi-Fi 6 is selected for medical tablets and laptops because clinical staff require mobility while maintaining secure access to hospital applications.

6.4 Implementation Evidence

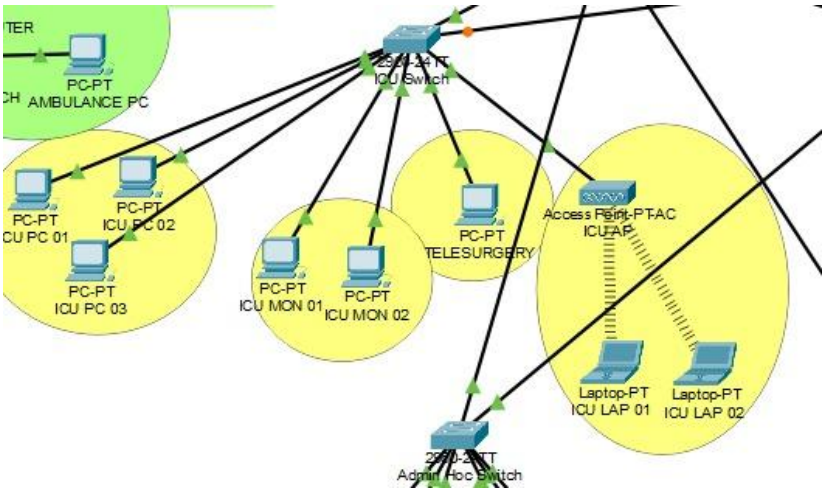


Figure 16 : ICU Ward Configuration

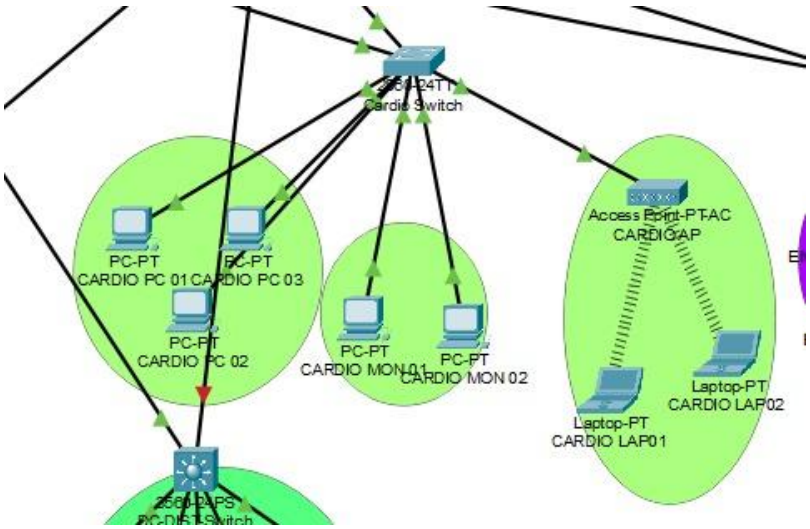


Figure 17: Cardio Ward Configuration

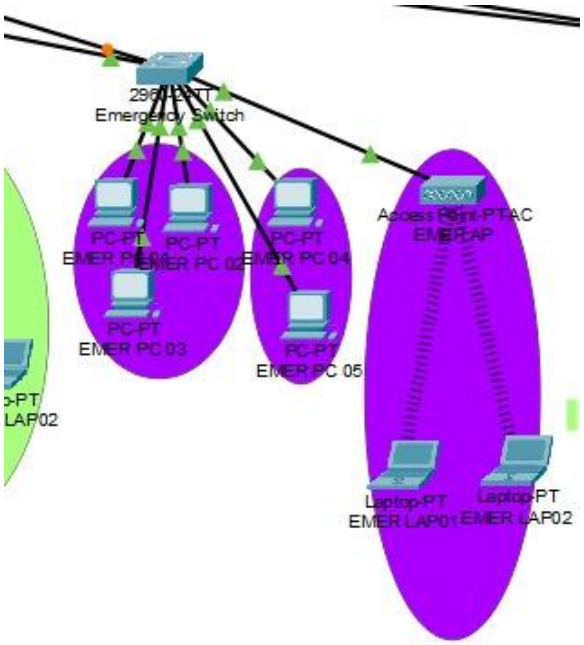


Figure 18 : Emergency Ward Configuration

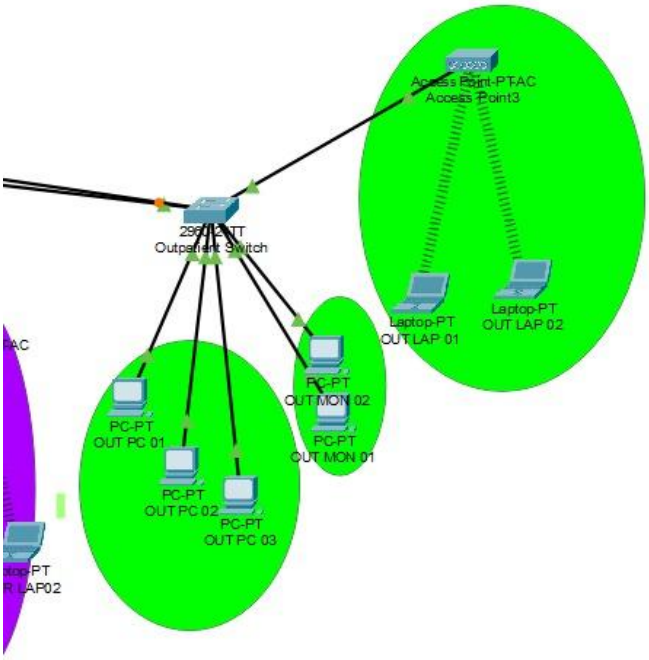


Figure 19 : Out Patient Configuration

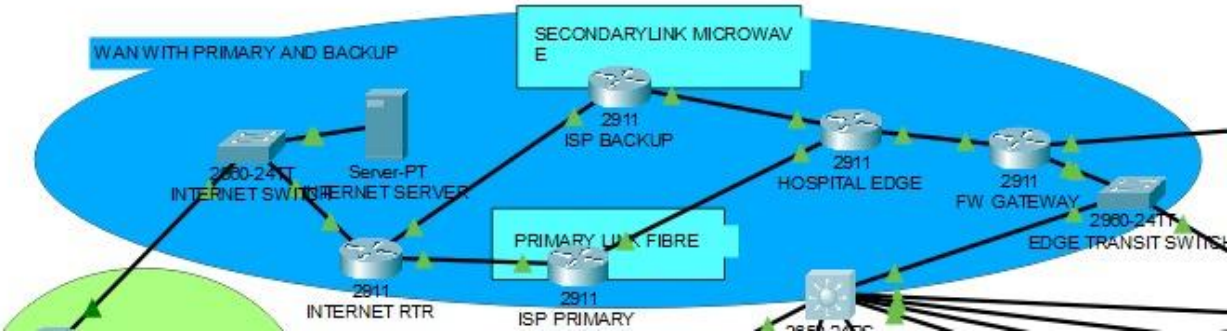


Figure 20 : WAN Configuration

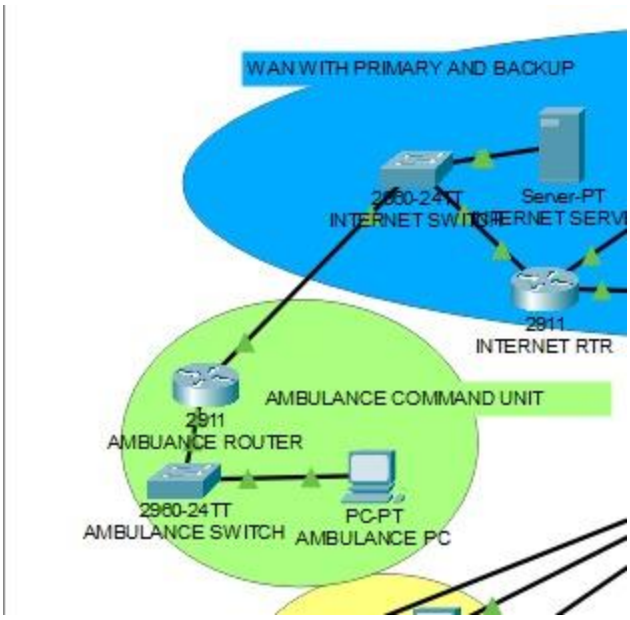


Figure 21 : Ambulance Configuration

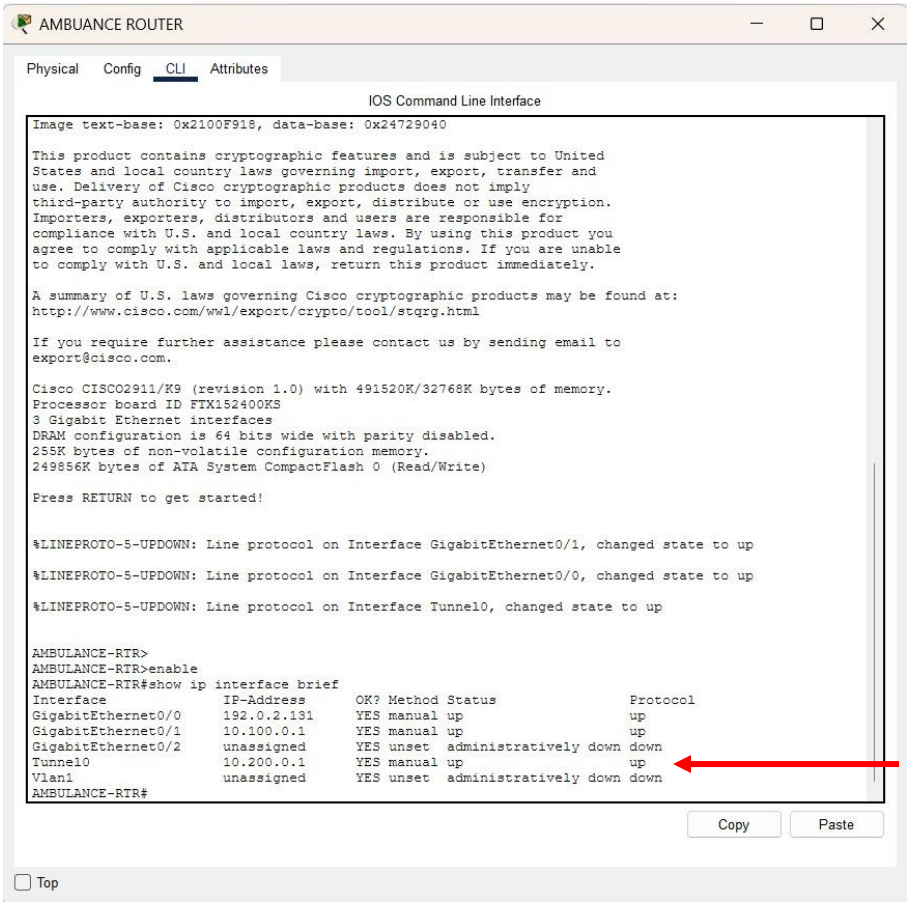


Figure 22 : Checking The External WAN Or Tunnel From Ambulance Router

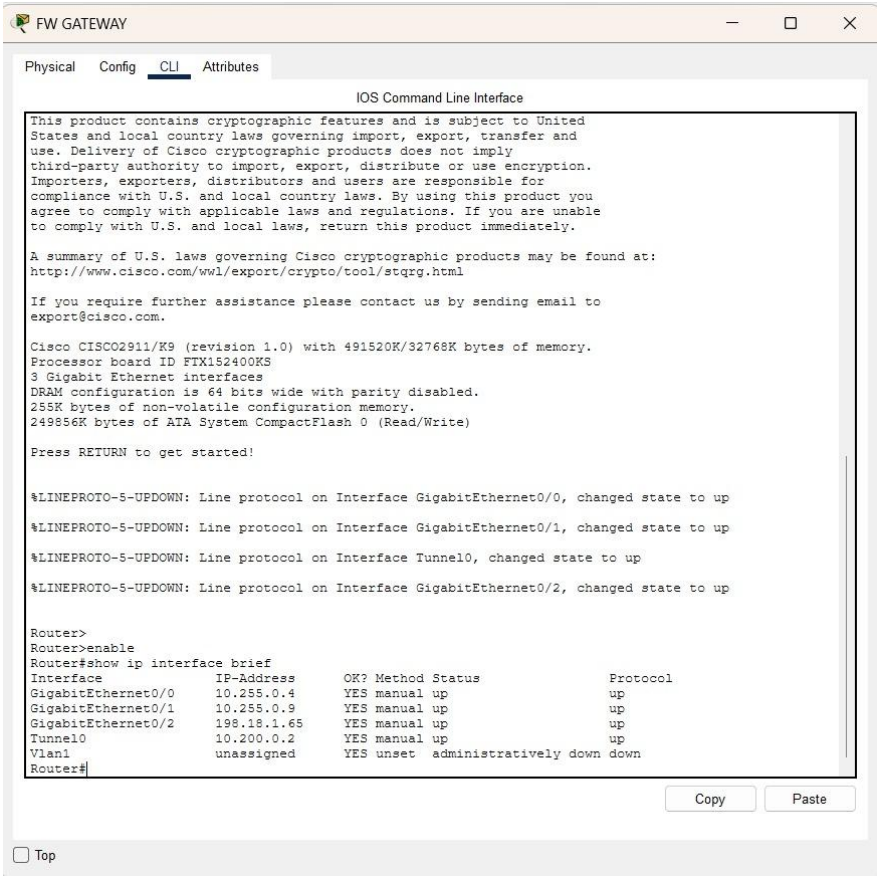


Figure 23 : Checking The External WAN Or Tunnel From FW Gateway

```
AMBULANCE-RTR#ping 10.200.0.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.200.0.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms

AMBULANCE-RTR#
```

Figure 24 : successful GRE Tunnel connectivity Test

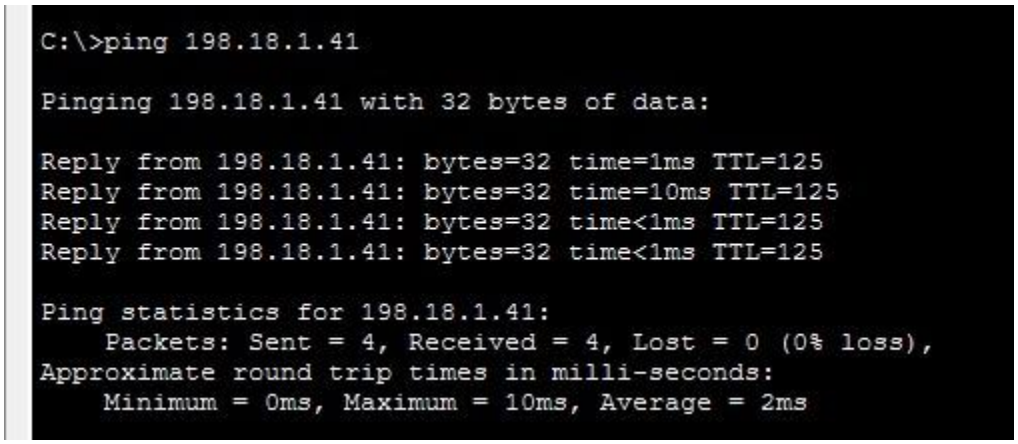


Figure 25 : Restricted Ambulance Access to Unauthorized Hospital Networks(PACS)

7. VLAN AND NETWORK SEGMENTATION DESIGN

7.1 VLAN Allocation

Table 13 : VLAN Allocation

VLAN	Department / Purpose	IPv4 Network	Default Gateway
10	ICU / Surgical	198.18.0.0/26	198.18.0.1
20	Cardiology	198.18.0.64/26	198.18.0.65
30	Emergency / Trauma	198.18.0.128/26	198.18.0.129
40	Outpatient Diagnostics	198.18.0.192/26	198.18.0.193
50	Administration	198.18.1.0/28	198.18.1.1
60	HOC	198.18.1.16/28	198.18.1.17
70	Centralized Data Centre	198.18.1.32/27	198.18.1.33
80	DMZ	198.18.1.64/28	198.18.1.65
90	Network Management	198.18.1.80/28	198.18.1.81
999	Core–Firewall Transit	10.255.0.0/29	—

```
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Gig1/0/9, Gig1/0/10, Gig1/0/11, Gig1/0/12
                                                Gig1/0/13, Gig1/0/14, Gig1/0/15, Gig1/0/16
                                                Gig1/0/17, Gig1/0/18, Gig1/0/19, Gig1/0/20
                                                Gig1/0/21, Gig1/0/22, Gig1/0/23, Gig1/0/24
                                                Gig1/1/1, Gig1/1/2, Gig1/1/3, Gig1/1/4
10   ICU-SURGICAL                     active
20   CARDIOLOGY                       active
30   EMERGENCY_TRAUMA                 active
40   OUTPATIENT                       active
50   ADMINISTRATION                   active
60   HOC                              active
70   SERVERS                          active
90   MANAGEMENT                       active
999  EDGE-TRANSIT                     active    Gig1/0/8
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#
```

Figure 26 : VLAN Configuration On CORE-SW1

```
1005 trnet-default                    active
Switch#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig1/0/1    on           802.1q         trunking      1
Gig1/0/2    on           802.1q         trunking      1
Gig1/0/3    on           802.1q         trunking      1
Gig1/0/4    on           802.1q         trunking      1
Gig1/0/5    on           802.1q         trunking      1
Gig1/0/6    on           802.1q         trunking      1
Gig1/0/7    on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gig1/0/1    10,90
Gig1/0/2    10,20,30,40,50,60,70,90
Gig1/0/3    20,90
Gig1/0/4    30,90
Gig1/0/5    40,90
Gig1/0/6    50,60,90
Gig1/0/7    70,90

Port        Vlans allowed and active in management domain
Gig1/0/1    10,90
Gig1/0/2    10,20,30,40,50,60,70,90
Gig1/0/3    20,90
Gig1/0/4    30,90
Gig1/0/5    40,90
Gig1/0/6    50,60,90
Gig1/0/7    70,90

Port        Vlans in spanning tree forwarding state and not pruned
Gig1/0/1    10,90
Gig1/0/2    10,20,30,40,50,60,70,90
Gig1/0/3    20,90
Gig1/0/4    30,90
Gig1/0/5    40,90
Gig1/0/6    50,60,90
Gig1/0/7    70,90

Switch#
Switch#
Switch#
Switch#
```

Figure 27 : VLAN Trunk Links Between the Core and Access Networks

- IEEE 802.1Q trunk links were used between the multilayer core and access switches to transport the required VLANs while maintaining logical separation between departments.

```
Switch#
Switch#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet1/0/1   unassigned      YES unset  up                    up
GigabitEthernet1/0/2   unassigned      YES unset  up                    up
GigabitEthernet1/0/3   unassigned      YES unset  up                    up
GigabitEthernet1/0/4   unassigned      YES unset  up                    up
GigabitEthernet1/0/5   unassigned      YES unset  up                    up
GigabitEthernet1/0/6   unassigned      YES unset  up                    up
GigabitEthernet1/0/7   unassigned      YES unset  up                    up
GigabitEthernet1/0/8   unassigned      YES unset  up                    up
GigabitEthernet1/0/9   unassigned      YES unset  down                  down
GigabitEthernet1/0/10  unassigned      YES unset  down                  down
GigabitEthernet1/0/11  unassigned      YES unset  down                  down
GigabitEthernet1/0/12  unassigned      YES unset  down                  down
GigabitEthernet1/0/13  unassigned      YES unset  down                  down
GigabitEthernet1/0/14  unassigned      YES unset  down                  down
GigabitEthernet1/0/15  unassigned      YES unset  down                  down
GigabitEthernet1/0/16  unassigned      YES unset  down                  down
GigabitEthernet1/0/17  unassigned      YES unset  down                  down
GigabitEthernet1/0/18  unassigned      YES unset  down                  down
GigabitEthernet1/0/19  unassigned      YES unset  down                  down
GigabitEthernet1/0/20  unassigned      YES unset  down                  down
GigabitEthernet1/0/21  unassigned      YES unset  down                  down
GigabitEthernet1/0/22  unassigned      YES unset  down                  down
GigabitEthernet1/0/23  unassigned      YES unset  down                  down
GigabitEthernet1/0/24  unassigned      YES unset  down                  down
GigabitEthernet1/1/1   unassigned      YES unset  down                  down
GigabitEthernet1/1/2   unassigned      YES unset  down                  down
GigabitEthernet1/1/3   unassigned      YES unset  down                  down
GigabitEthernet1/1/4   unassigned      YES unset  down                  down
Vlan1                  unassigned      YES unset  administratively down down
Vlan10                 198.18.0.2      YES manual up                    up
Vlan20                 198.18.0.66     YES manual up                    up
Vlan30                 198.18.0.130    YES manual up                    up
Vlan40                 198.18.0.194    YES manual up                    up
Vlan50                 198.18.1.2      YES manual up                    up
Vlan60                 198.18.1.18     YES manual up                    up
Vlan70                 198.18.1.34     YES manual up                    up
Vlan90                 198.18.1.82     YES manual up                    up
Vlan999                10.255.0.2      YES manual up                    up
Switch#
```

Figure 28 : Layer 3 SVI Interfaces Configured on the Hospital Core

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.....: FE80::201:96FF:FE55:C732
   IPv6 Address.....: 2001:DB8:ABCD:10:201:96FF:FE55:C732
   IPv4 Address.....: 198.18.0.14
   Subnet Mask.....: 255.255.255.192
   Default Gateway.....: FE80::206:2AFF:FED5:E401
                         198.18.0.1

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.....: ::
   IPv6 Address.....: ::
   IPv4 Address.....: 0.0.0.0
   Subnet Mask.....: 0.0.0.0
   Default Gateway.....: ::
                         0.0.0.0

C:\>ping 198.18.1.41

Pinging 198.18.1.41 with 32 bytes of data:

Reply from 198.18.1.41: bytes=32 time<1ms TTL=127
Reply from 198.18.1.41: bytes=32 time<1ms TTL=127
Reply from 198.18.1.41: bytes=32 time<1ms TTL=127
Reply from 198.18.1.41: bytes=32 time<1ms TTL=127

Ping statistics for 198.18.1.41:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Figure 29 : ICU VLAN Addressing and Successful Inter-VLAN Connectivity to the EMR Server

```
Switch>
Switch>
Switch>enable
Switch#show vlan brief

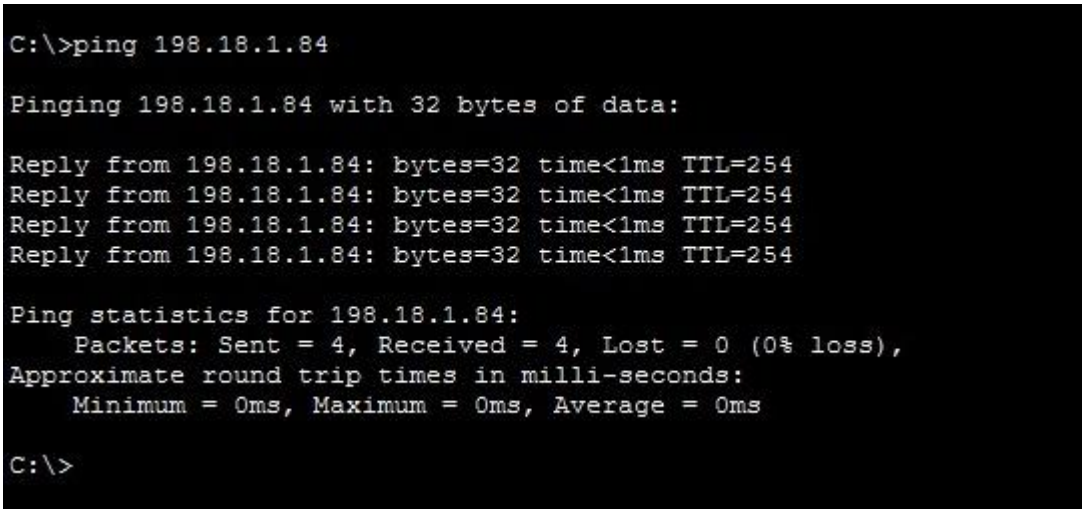
VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/3, Fa0/4, Fa0/5, Fa0/6
                                                Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                                Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Fa0/23, Fa0/24, Gig0/1, Gig0/2
80   DMZ                              active    Fa0/1, Fa0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#
```

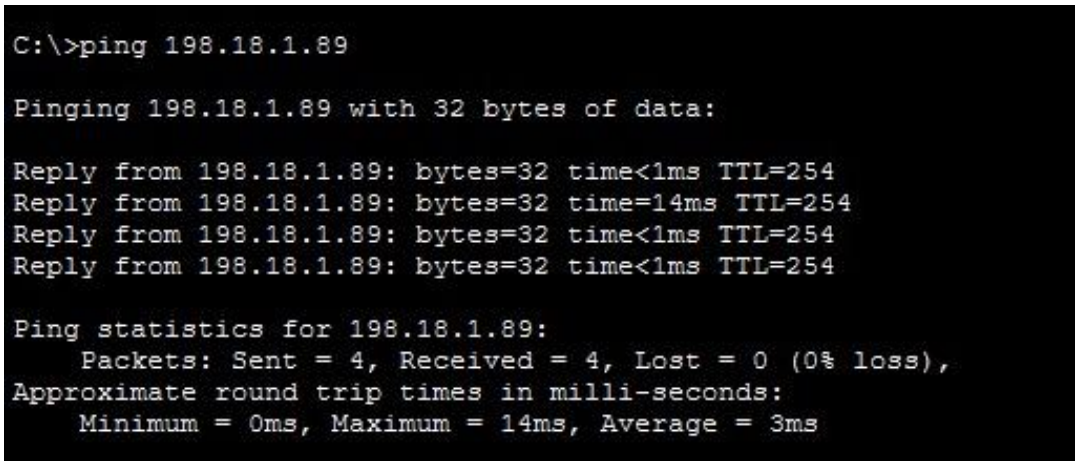
Figure 30 : Dedicated VLAN 80 DMZ Network

7.2 Management VLAN

The network-management interfaces were placed in VLAN 90:

- Virtual Gateway = 198.18.1.81
- CORE-SW1 = 198.18.1.82
- CORE-SW2 = 198.18.1.83
- ICU-SW = 198.18.1.84
- CARDIO-SW = 198.18.1.85
- EMERGENCY-SW = 198.18.1.86
- OUTPATIENT-SW = 198.18.1.87
- ADMIN-HOC-SW = 198.18.1.88
- DC-DIST-SW = 198.18.1.89





8. IPV4 AND IPV6 ADDRESSING DESIGN

8.1 IPv4 Configuration Evidence

```
Switch#enable
Switch#show ip interface brief
 %HSRP-6-STATECHANGE: Vlan30 Grp 30 state Standby -> Active

Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet1/0/1   unassigned      YES unset  up                    up
GigabitEthernet1/0/2   unassigned      YES unset  up                    up
GigabitEthernet1/0/3   unassigned      YES unset  up                    up
GigabitEthernet1/0/4   unassigned      YES unset  up                    up
GigabitEthernet1/0/5   unassigned      YES unset  up                    up
GigabitEthernet1/0/6   unassigned      YES unset  up                    up
GigabitEthernet1/0/7   unassigned      YES unset  up                    up
GigabitEthernet1/0/8   unassigned      YES unset  up                    up
GigabitEthernet1/0/9   unassigned      YES unset  down                  down
GigabitEthernet1/0/10  unassigned      YES unset  down                  down
GigabitEthernet1/0/11  unassigned      YES unset  down                  down
GigabitEthernet1/0/12  unassigned      YES unset  down                  down
GigabitEthernet1/0/13  unassigned      YES unset  down                  down
GigabitEthernet1/0/14  unassigned      YES unset  down                  down
GigabitEthernet1/0/15  unassigned      YES unset  down                  down
GigabitEthernet1/0/16  unassigned      YES unset  down                  down
GigabitEthernet1/0/17  unassigned      YES unset  down                  down
GigabitEthernet1/0/18  unassigned      YES unset  down                  down
GigabitEthernet1/0/19  unassigned      YES unset  down                  down
GigabitEthernet1/0/20  unassigned      YES unset  down                  down
GigabitEthernet1/0/21  unassigned      YES unset  down                  down
GigabitEthernet1/0/22  unassigned      YES unset  down                  down
GigabitEthernet1/0/23  unassigned      YES unset  down                  down
GigabitEthernet1/0/24  unassigned      YES unset  down                  down
GigabitEthernet1/1/1   unassigned      YES unset  down                  down
 --More--
 %HSRP-6-STATECHANGE: Vlan60 Grp 60 state Standby -> Active
GigabitEthernet1/1/2   unassigned      YES unset  down                  down
GigabitEthernet1/1/3   unassigned      YES unset  down                  down
GigabitEthernet1/1/4   unassigned      YES unset  down                  down
Vlan1                  unassigned      YES unset  administratively down down
Vlan10                 198.18.0.2      YES manual up                    up
Vlan20                 198.18.0.66     YES manual up                    up
Vlan30                 198.18.0.130    YES manual up                    up
Vlan40                 198.18.0.194    YES manual up                    up
 --More--
 %HSRP-6-STATECHANGE: Vlan70 Grp 70 state Standby -> Active
Vlan50                 198.18.1.2      YES manual up                    up
Vlan60                 198.18.1.18     YES manual up                    up
Vlan70                 198.18.1.34     YES manual up                    up
Vlan90                 198.18.1.82     YES manual up                    up
 --More--
 %HSRP-6-STATECHANGE: Vlan999 Grp 99 state Standby -> Active

 %HSRP-6-STATECHANGE: Vlan50 Grp 50 state Standby -> Active
Vlan999                10.255.0.2      YES manual up                    up
```

Figure 31 : IPv4 Layer 3 Interfaces Configured on CORE-SW1

8.2 IPv6 Configuration Evidence

```
Switch#show ipv6 interface brief
GigabitEthernet1/0/1       [up/up]
    unassigned
GigabitEthernet1/0/2       [up/up]
    unassigned
GigabitEthernet1/0/3       [up/up]
    unassigned
GigabitEthernet1/0/4       [up/up]
    unassigned
GigabitEthernet1/0/5       [up/up]
    unassigned
GigabitEthernet1/0/6       [up/up]
    unassigned
GigabitEthernet1/0/7       [up/up]
    unassigned
GigabitEthernet1/0/8       [up/up]
    unassigned
GigabitEthernet1/0/9       [down/down]
    unassigned
GigabitEthernet1/0/10      [down/down]
    unassigned
GigabitEthernet1/0/11      [down/down]
    unassigned
GigabitEthernet1/0/12      [down/down]
    unassigned
GigabitEthernet1/0/13      [down/down]
    unassigned
GigabitEthernet1/0/14      [down/down]
    unassigned
GigabitEthernet1/0/15      [down/down]
    unassigned
GigabitEthernet1/0/16      [down/down]
    unassigned
GigabitEthernet1/0/17      [down/down]
    unassigned
GigabitEthernet1/0/18      [down/down]
    unassigned
GigabitEthernet1/0/19      [down/down]
    unassigned
GigabitEthernet1/0/20      [down/down]
    unassigned
GigabitEthernet1/0/21      [down/down]
    unassigned
GigabitEthernet1/0/22      [down/down]
    unassigned
GigabitEthernet1/0/23      [down/down]
    unassigned
GigabitEthernet1/0/24      [down/down]
    unassigned
GigabitEthernet1/1/1       [down/down]
    unassigned
GigabitEthernet1/1/2       [down/down]
    unassigned
GigabitEthernet1/1/3       [down/down]
    unassigned
    unassigned
Vlan1                      [administratively down/down]
    unassigned
Vlan10                     [up/up]
    FE80::206:2AFF:FED5:E401
    2001:DB8:ABCD:10::2
Vlan20                     [up/up]
    FE80::206:2AFF:FED5:E402
    2001:DB8:ABCD:20::2
Vlan30                     [up/up]
    FE80::206:2AFF:FED5:E403
    2001:DB8:ABCD:30::2
Vlan40                     [up/up]
    FE80::206:2AFF:FED5:E404
    2001:DB8:ABCD:40::2
Vlan50                     [up/up]
    FE80::206:2AFF:FED5:E405
    2001:DB8:ABCD:50::2
Vlan60                     [up/up]
    FE80::206:2AFF:FED5:E406
    2001:DB8:ABCD:60::2
Vlan70                     [up/up]
    FE80::206:2AFF:FED5:E407
    2001:DB8:ABCD:70::2
Vlan90                     [up/up]
    FE80::206:2AFF:FED5:E408
    2001:DB8:ABCD:90::2
```

Figure 32 : IPv6 Interfaces Configured on CORE-SW1

8.3 Dual-Stack Client Verification

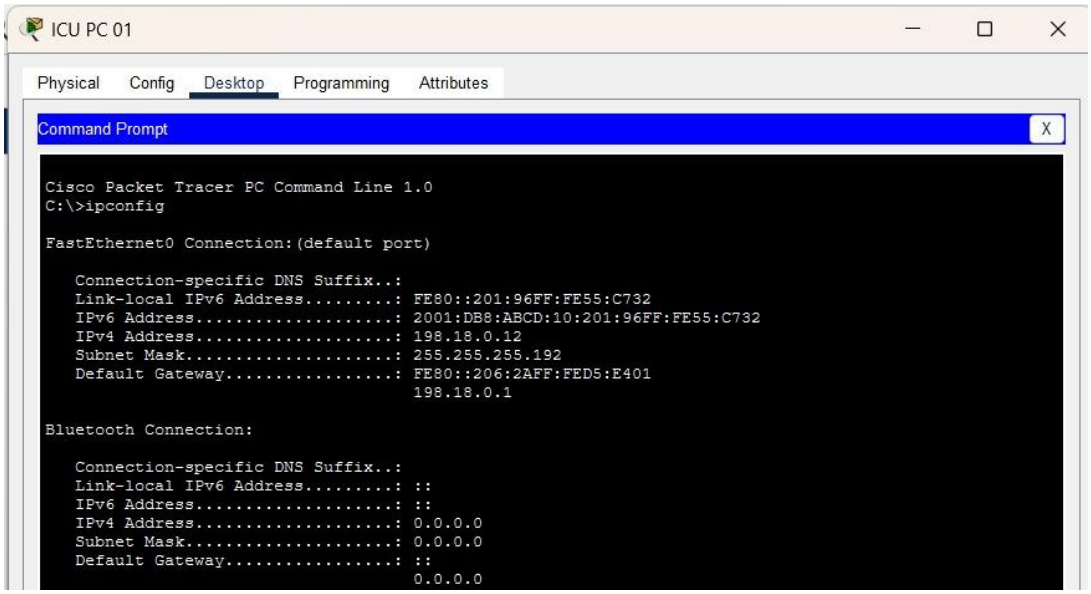


Figure 33 : Dual-Stack IPv4 and IPv6 Configuration on an ICU Workstation

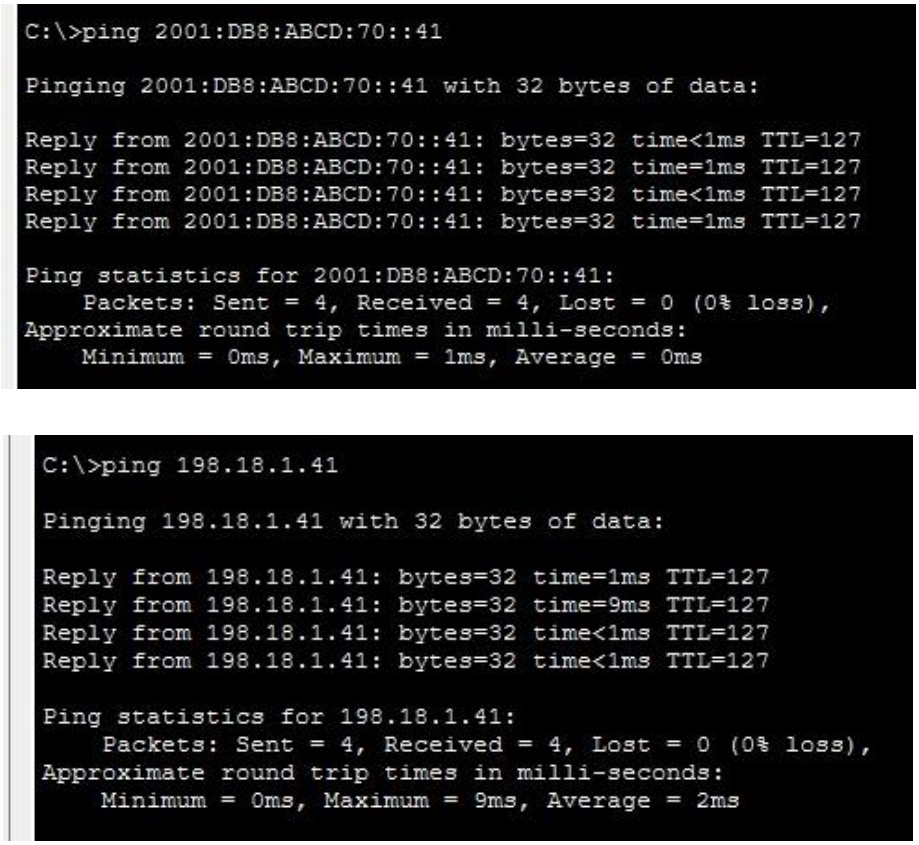


Figure 34 : Successful IPv4 and IPv6 Communication Between ICU and EMR Server

9. ROUTING PROTOCOL AND ROUTING DESIGN

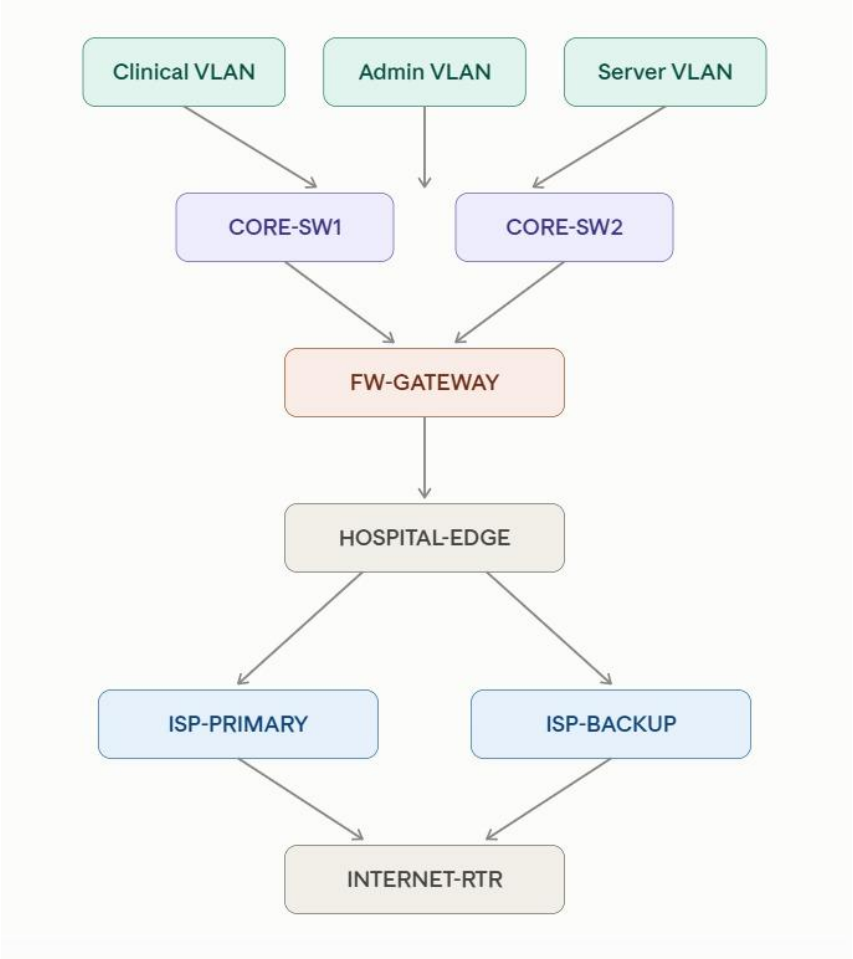


Figure 35 : Routing Architecture

The multilayer core switches perform routing between internal hospital VLANs. Static routes are used between the core and firewall because this is a small, controlled transit segment. OSPF is used across the external redundant WAN infrastructure to provide dynamic route exchange and automatic path selection.

9.1 Static Routing Between Core and Firewall

The core switches use the firewall as the next hop for destinations outside the hospital LAN:

Default route → 10.255.0.4

10.255.0.4 is the inside/transit address of FW-GATEWAY.

The firewall uses:

Hospital network:

198.18.0.0/23 → 10.255.0.1

Default route:

0.0.0.0/0 → 10.255.0.10

where:

10.255.0.1 = Core HSRP virtual address

10.255.0.10 = HOSPITAL-EDGE

```
Switch#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.255.0.4 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.255.0.0/29 is directly connected, Vlan999
L       10.255.0.2/32 is directly connected, Vlan999
     198.18.0.0/24 is variably subnetted, 8 subnets, 2 masks
C       198.18.0.0/26 is directly connected, Vlan10
L       198.18.0.2/32 is directly connected, Vlan10
C       198.18.0.64/26 is directly connected, Vlan20
L       198.18.0.66/32 is directly connected, Vlan20
C       198.18.0.128/26 is directly connected, Vlan30
L       198.18.0.130/32 is directly connected, Vlan30
C       198.18.0.192/26 is directly connected, Vlan40
L       198.18.0.194/32 is directly connected, Vlan40
     198.18.1.0/24 is variably subnetted, 8 subnets, 3 masks
C       198.18.1.0/28 is directly connected, Vlan50
L       198.18.1.2/32 is directly connected, Vlan50
C       198.18.1.16/28 is directly connected, Vlan60
L       198.18.1.18/32 is directly connected, Vlan60
C       198.18.1.32/27 is directly connected, Vlan70
L       198.18.1.34/32 is directly connected, Vlan70
C       198.18.1.80/28 is directly connected, Vlan90
L       198.18.1.82/32 is directly connected, Vlan90
S*   0.0.0.0/0 [1/0] via 10.255.0.4
```

Figure 36 : CORE-SW1 Routing Table Showing Internal Networks and Default Route

```
Router>
Router>enable
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.255.0.10 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 7 subnets, 4 masks
S       10.100.0.0/24 [1/0] via 10.200.0.1
C       10.200.0.0/30 is directly connected, Tunnel0
L       10.200.0.2/32 is directly connected, Tunnel0
C       10.255.0.0/29 is directly connected, GigabitEthernet0/0
L       10.255.0.4/32 is directly connected, GigabitEthernet0/0
C       10.255.0.8/30 is directly connected, GigabitEthernet0/1
L       10.255.0.9/32 is directly connected, GigabitEthernet0/1
S    198.18.0.0/23 [1/0] via 10.255.0.1
     198.18.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       198.18.1.64/28 is directly connected, GigabitEthernet0/2
L       198.18.1.65/32 is directly connected, GigabitEthernet0/2
S*   0.0.0.0/0 [1/0] via 10.255.0.10
```

Figure 37 : Firewall Gateway Routing Table

9.2 OSPF WAN Routing

OSPF Area 0 was configured across the hospital WAN infrastructure to exchange routes between:

- HOSPITAL-EDGE
- ISP-PRIMARY
- ISP-BACKUP
- INTERNET-RTR

Table 14 : The primary and secondary paths use different OSPF costs

WAN Path	OSPF Cost	Function
Primary fibre	10	Preferred path
Secondary microwave	100	Backup path

```
Router>
Router>enable
Router#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
203.0.113.2       1   FULL/DR         00:00:30    203.0.113.2     GigabitEthernet0/1
198.51.100.2      1   FULL/BDR        00:00:30    198.51.100.2    GigabitEthernet0/2
Router#
```

Figure 38 : OSPF Neighbour Relationships On HOSPITAL-EDGE

9.3 Route Selection Verification

```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.255.0.8/30 is directly connected, GigabitEthernet0/0
L       10.255.0.10/32 is directly connected, GigabitEthernet0/0
     192.0.2.0/24 is variably subnetted, 3 subnets, 2 masks
O       192.0.2.0/30 [110/11] via 203.0.113.2, 00:18:38, GigabitEthernet0/1
O       192.0.2.4/30 [110/101] via 198.51.100.2, 00:18:48, GigabitEthernet0/2
O       192.0.2.128/25 [110/12] via 203.0.113.2, 00:18:38, GigabitEthernet0/1
S    198.18.0.0/23 [1/0] via 10.255.0.9
     198.51.100.0/24 is variably subnetted, 2 subnets, 2 masks
C       198.51.100.0/30 is directly connected, GigabitEthernet0/2
L       198.51.100.1/32 is directly connected, GigabitEthernet0/2
     203.0.113.0/24 is variably subnetted, 2 subnets, 2 masks
C       203.0.113.0/30 is directly connected, GigabitEthernet0/1
L       203.0.113.1/32 is directly connected, GigabitEthernet0/1
```

Figure 39 : OSPF Route Selection Toward the External Network

9.4 Ambulance Routing

The remote ambulance uses a specific route through the GRE tunnel to reach the hospital network:

198.18.0.0/23

↓

10.200.0.2

The firewall has the return route:

10.100.0.0/24

↓

10.200.0.1

```
AMBULANCE-RTR>enable
AMBULANCE-RTR#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 192.0.2.129 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 4 subnets, 3 masks
C       10.100.0.0/24 is directly connected, GigabitEthernet0/1
L       10.100.0.1/32 is directly connected, GigabitEthernet0/1
C       10.200.0.0/30 is directly connected, Tunnel0
L       10.200.0.1/32 is directly connected, Tunnel0
     192.0.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.0.2.128/25 is directly connected, GigabitEthernet0/0
L       192.0.2.131/32 is directly connected, GigabitEthernet0/0
S    198.18.0.0/23 [1/0] via 10.200.0.2
S*   0.0.0.0/0 [1/0] via 192.0.2.129
```

Figure 40 : Ambulance Router Route Toward The Hospital Through The GRE Tunnel

10. CORE INFRASTRUCTURE FOR REAL-TIME TELESURGERY AND PATIENT TELEMETRY

10.1 Critical Medical Service Architecture

The hospital network was designed to support two major real-time healthcare applications:

- ✓ Robotic telesurgery traffic
- ✓ Continuous patient telemetry

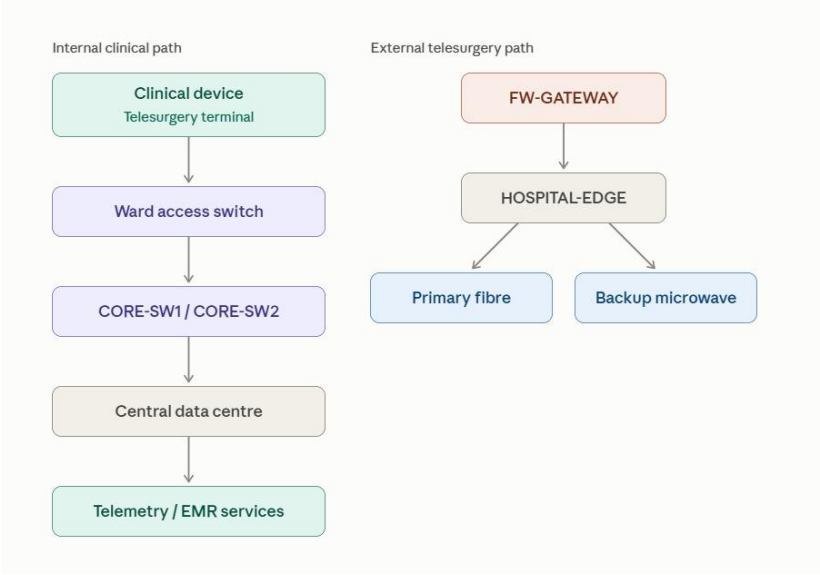


Figure 41 : Critical Medical Service Architecture Communication Path

10.2 Telesurgery Infrastructure

A dedicated endpoint was configured to represent the robotic telesurgery system:

TELESURGERY IP Address: 198.18.0.30

The endpoint is connected through the hospital core and can reach external healthcare infrastructure using the primary fibre WAN connection. If the primary connection fails, the backup microwave path provides alternative connectivity.

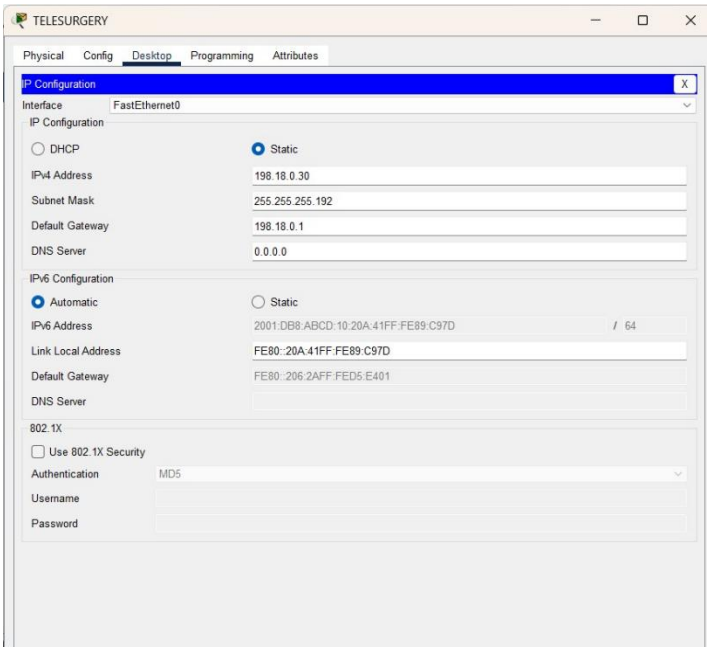


Figure 42 : IP Configuration of the Robotic Telesurgery Endpoint

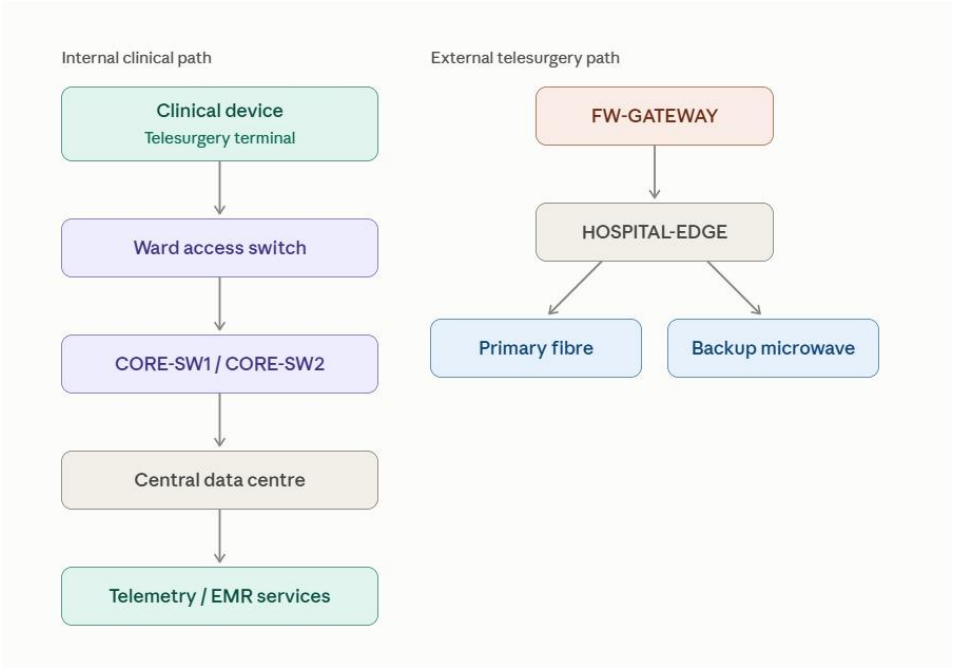


Figure 43: critical medical path service architecture

10.3 Patient Telemetry Infrastructure

Patient-monitoring devices in the clinical wards communicate with the centralized Telemetry server:

Telemetry Server 198.18.1.44

The server is located in the centralized Data Centre within VLAN 70.

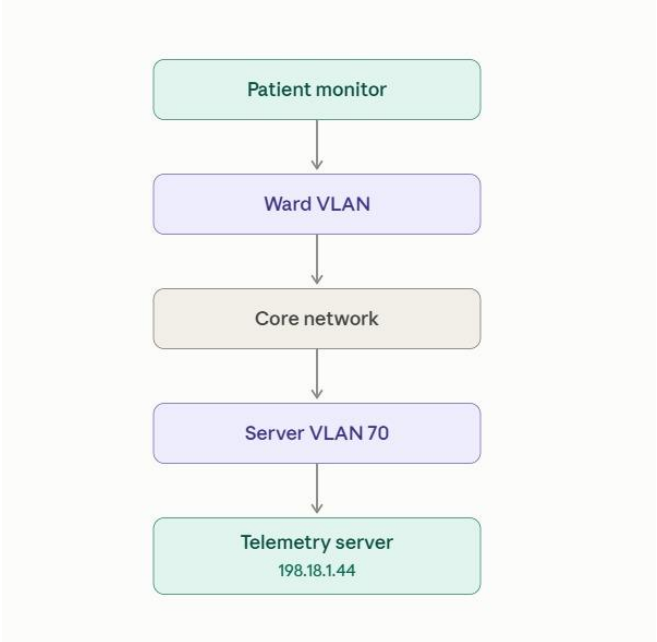


Figure 44 : Patient Telemetry Infrastructure communication path

10.4 Telemetry Connectivity Verification

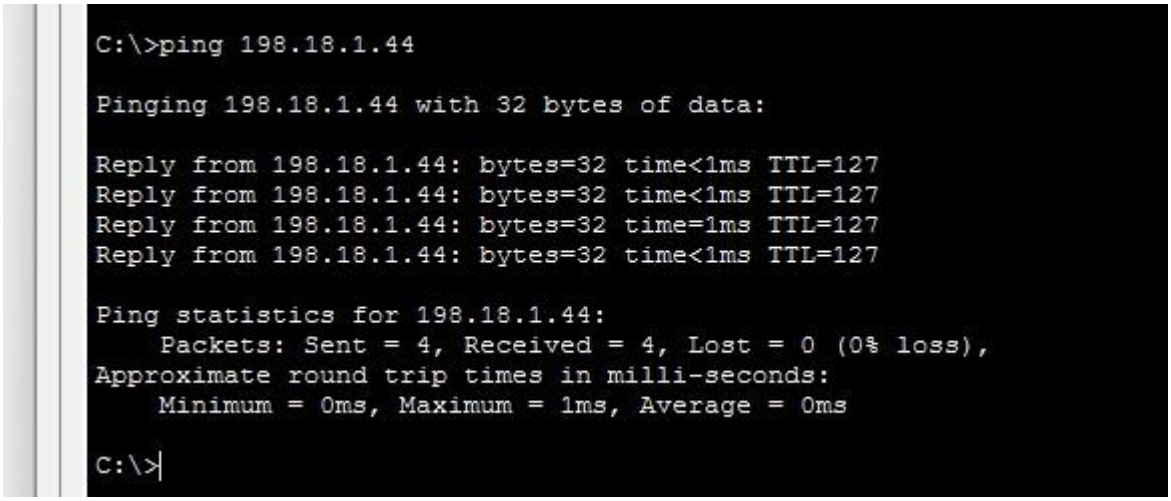


Figure 45 : Successful Connectivity Between a Clinical Device and the Telemetry Server

10.5 Infrastructure Supporting Critical Traffic

Table 15 : Infrastructure Supporting Critical Traffic

Requirement	Implemented Infrastructure
Low-latency telesurgery	Dedicated primary fibre WAN
WAN availability	Secondary microwave link
Core availability	Redundant CORE-SW1 and CORE-SW2
Gateway redundancy	HSRP
Patient telemetry collection	Telemetry server 198.18.1.44
Telesurgery endpoint	198.18.0.30
Traffic prioritization	QoS policy
Automatic WAN path recovery	OSPF failover
Medical server connectivity	Centralized Data Centre

11. QUALITY OF SERVICE (QOS) IMPLEMENTATION AND VERIFICATION

11.1 QoS Requirement

The hospital network carries traffic with different levels of importance. Real-time robotic telesurgery and patient telemetry require preferential treatment, while PACS and EMR traffic also need sufficient bandwidth during congestion.

11.2 QoS Classes Implemented

Table 16 : QoS Classes Implemented

Traffic Class	Purpose	QoS Treatment
TELESURGERY	Robotic telesurgery traffic	Strict priority, 30%
TELEMETRY	Patient-monitoring traffic	10% bandwidth
EMR_TRAFFIC	EMR/EHR access	15% bandwidth
PACS_TRAFFIC	Medical imaging traffic	20% bandwidth
class-default	Other traffic	Remaining bandwidth / fair queue

\

11.3 QoS Policy Deployment

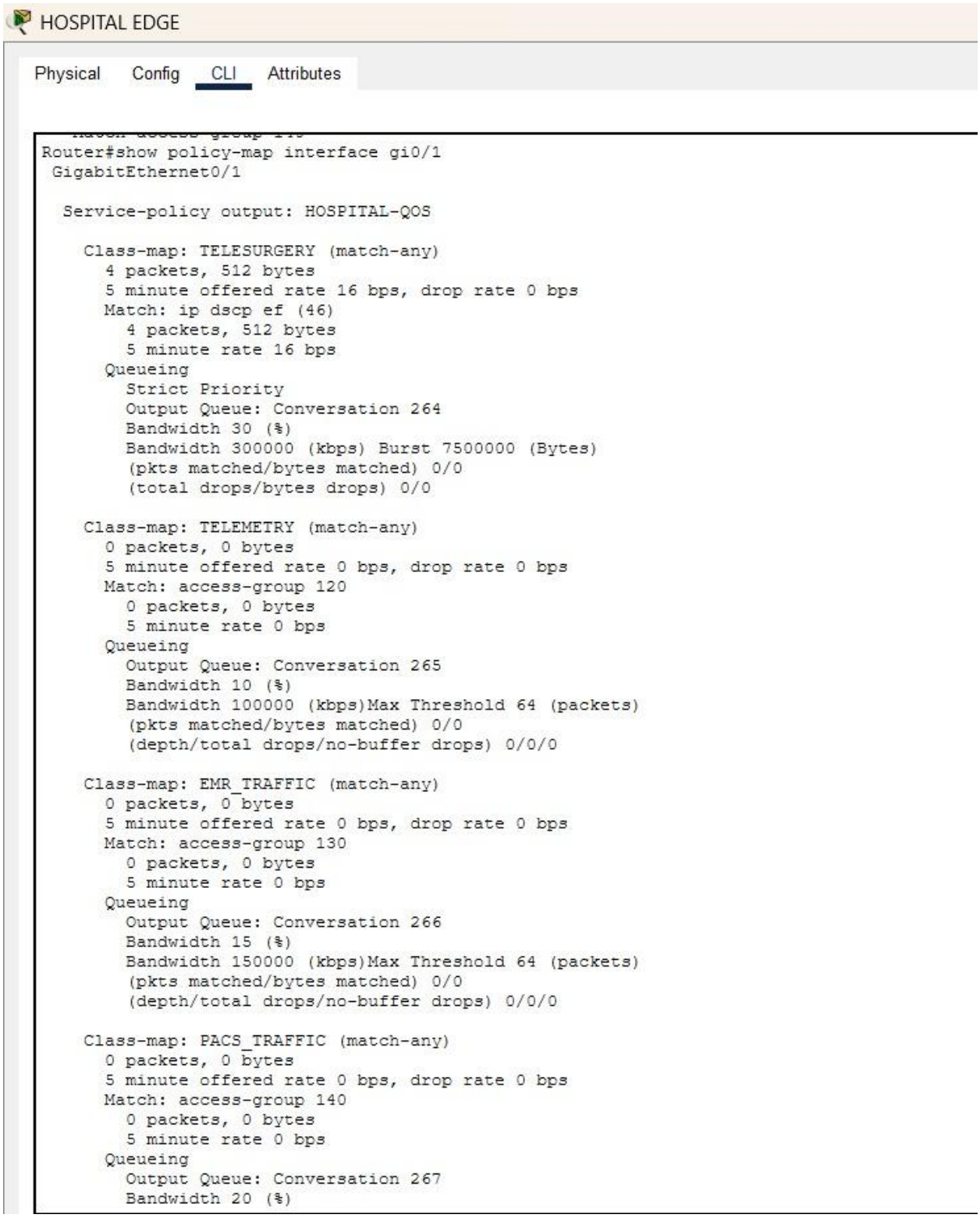


Figure 46 : HOSPITAL-QOS Policy Applied to the Primary WAN Interface

11.4 Telesurgery Traffic Verification

HOSPITAL EDGE

Physical Config CLI Attributes

```
Router#show policy-map interface gi0/1
 GigabitEthernet0/1

  Service-policy output: HOSPITAL-QOS

    Class-map: TELESURGERY (match-any)
      12 packets, 1536 bytes
      5 minute offered rate 41 bps, drop rate 0 bps
      Match: ip dscp ef (46)
        12 packets, 1536 bytes
        5 minute rate 41 bps
      Queueing
        Strict Priority
        Output Queue: Conversation 264
        Bandwidth 30 (%)
        Bandwidth 300000 (kbps) Burst 7500000 (Bytes)
        (pkts matched/bytes matched) 0/0
        (total drops/bytes drops) 0/0

    Class-map: TELEMETRY (match-any)
      0 packets, 0 bytes
      5 minute offered rate 0 bps, drop rate 0 bps
      Match: access-group 120
        0 packets, 0 bytes
        5 minute rate 0 bps
      Queueing
        Output Queue: Conversation 265
        Bandwidth 10 (%)
        Bandwidth 100000 (kbps)Max Threshold 64 (packets)
        (pkts matched/bytes matched) 0/0
        (depth/total drops/no-buffer drops) 0/0/0

    Class-map: EMR_TRAFFIC (match-any)
      0 packets, 0 bytes
      5 minute offered rate 0 bps, drop rate 0 bps
      Match: access-group 130
        0 packets, 0 bytes
        5 minute rate 0 bps
      Queueing
        Output Queue: Conversation 266
        Bandwidth 15 (%)
        Bandwidth 150000 (kbps)Max Threshold 64 (packets)
        (pkts matched/bytes matched) 0/0
        (depth/total drops/no-buffer drops) 0/0/0

    Class-map: PACS_TRAFFIC (match-any)
      0 packets, 0 bytes
      5 minute offered rate 0 bps, drop rate 0 bps
      Match: access-group 140
        0 packets, 0 bytes
        5 minute rate 0 bps
      Queueing
        Output Queue: Conversation 267
        Bandwidth 20 (%)
```

Figure 47 : Telesurgery Traffic Classified into the Strict-Priority QoS Class

12. FIREWALL, NAT AND SECURITY POLICIES

12.1 Firewall Gateway

FW-GATEWAY acts as the security boundary between the internal hospital network, DMZ, WAN, and remote ambulance connection. Its main security functions are:

- NAT/PAT for internal users
- Administration Internet restriction
- DMZ filtering
- Ambulance access control
- Separation between trusted and untrusted networks

12.2 NAT/PAT Implementation

```
Router#show ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
icmp 10.255.0.9:17     198.18.0.30:17     192.0.2.130:17     192.0.2.130:17
icmp 10.255.0.9:18     198.18.0.30:18     192.0.2.130:18     192.0.2.130:18
icmp 10.255.0.9:19     198.18.0.30:19     192.0.2.130:19     192.0.2.130:19
icmp 10.255.0.9:20     198.18.0.30:20     192.0.2.130:20     192.0.2.130:20
```

```
Router#show ip nat statistics
Total translations: 0 (0 static, 0 dynamic, 0 extended)
Outside Interfaces: GigabitEthernet0/1
Inside Interfaces: GigabitEthernet0/0
Hits: 13  Misses: 15
Expired translations: 15
Dynamic mappings:
Router#show access-lists 1
Standard IP access list 1
    permit 198.18.0.0 0.0.1.255 (30 match(es))
```

Figure 48 : NAT/PAT Translation on FW-GATEWAY

12.3 Administration Internet Restriction

Administration PCs are allowed to access local Billing and Inventory systems but are blocked from direct Internet access.

Policy:

Administration → Billing ☑

Administration → Inventory ☑

Administration → Internet ✗

```
Router#show access-lists OUTBOUND-POLICY
Extended IP access list OUTBOUND-POLICY
    deny ip 198.18.1.0 0.0.0.15 any
    permit ip any any (3316 match(es))

Router#
```

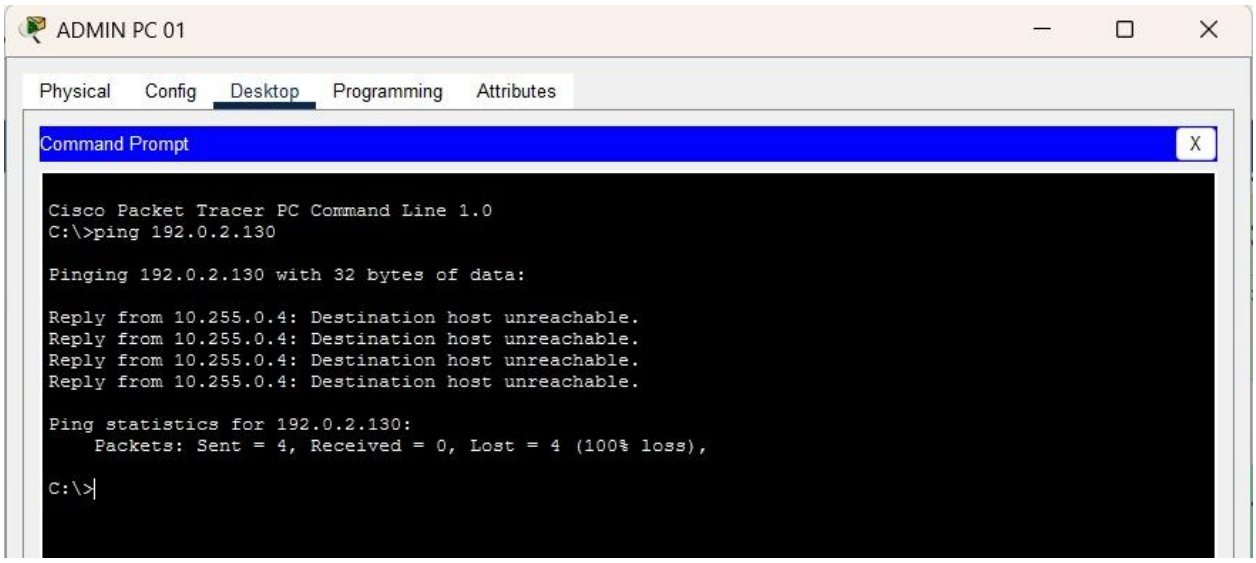


Figure 49 : Firewall ACL Restricting Administration Internet Access

12.4 Management Network Protection

Management access is restricted to the HOC network.

Expected policy:

HOC → Management ☑

ICU → Management ✗

Admin → Management ✗

```
Switch#show access-lists MGMT-ACCESS
Extended IP access list MGMT-ACCESS
    permit ip 198.18.1.16 0.0.0.15 198.18.1.80 0.0.0.15
    deny ip any 198.18.1.80 0.0.0.15
    permit ip any any
```

Figure 50 : Restricted Access to the Management VLAN

12.5 DMZ Security

The public web server is located at:

- DMZ-WEB-SERVER = 198.18.1.66
- Gateway = 198.18.1.65

The implemented policy is:

Internet → DMZ HTTP/HTTPS ☑

DMZ → EMR ✗

DMZ → PACS ✗

DMZ → Management ✗

```
Router#show access-lists DMZ-TO-INTERNAL
Extended IP access list DMZ-TO-INTERNAL
    deny ip 198.18.1.64 0.0.0.15 198.18.0.0 0.0.1.255
    permit ip any any

Router#show access-lists TO-DMZ
Extended IP access list TO-DMZ
    permit tcp any host 198.18.1.66 eq www
    permit tcp any host 198.18.1.66 eq 443
    permit icmp host 192.0.2.130 host 198.18.1.66
    permit ip 198.18.0.0 0.0.1.255 198.18.1.64 0.0.0.15
    deny ip any 198.18.1.64 0.0.0.15
    permit ip any any
```

Figure 51 : Successful Internet Access to the DMZ Web Server and DMZ Firewall ACLs Protecting Internal Hospital Systems

12.6 Ambulance Access Control

The ambulance is restricted to approved clinical services.

Policy:

Ambulance → EMR ☑

Ambulance → Telemetry ☑

Ambulance → PACS ✗

Ambulance → HOC ✗

Ambulance → Management ✗

```
Router#show access-lists AMBULANCE-VPN
Extended IP access list AMBULANCE-VPN
    permit ip 10.100.0.0 0.0.0.255 host 198.18.1.41
    permit ip 10.100.0.0 0.0.0.255 host 198.18.1.44
    deny ip 10.100.0.0 0.0.0.255 198.18.0.0 0.0.1.255
    permit ip any any (37 match(es))

Router#
```

Figure 52: ACL Restricting Remote Ambulance Access

13. DMZ IMPLEMENTATION AND VERIFICATION

13.1 DMZ Design

A separate DMZ was implemented for public-facing hospital services so that external users do not directly access sensitive internal systems.

Table 17 : DMZ addressing

Item	Configuration
DMZ Network	198.18.1.64/28
Subnet Mask	255.255.255.240
FW-GATEWAY Gi0/2	198.18.1.65
DMZ Web Server	198.18.1.66
DMZ VLAN	VLAN 80

13.2 External Web Access Verification

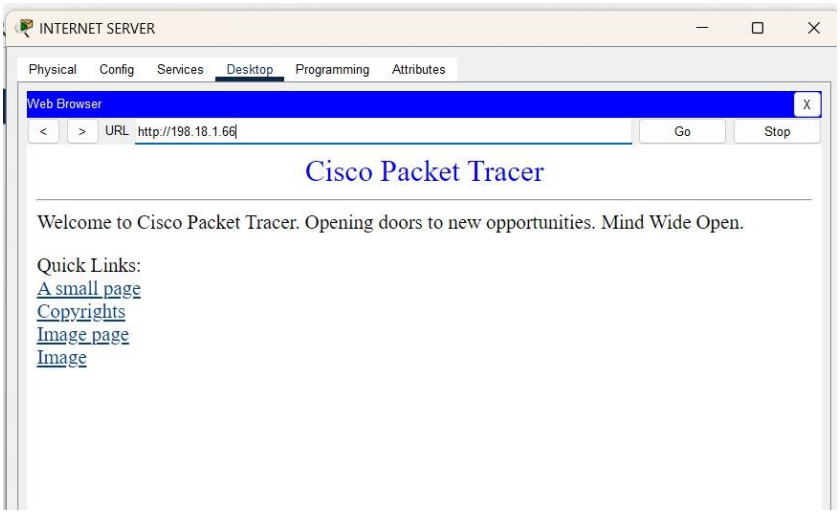


Figure 53 : Successful Internet Access to the Public DMZ Web Server

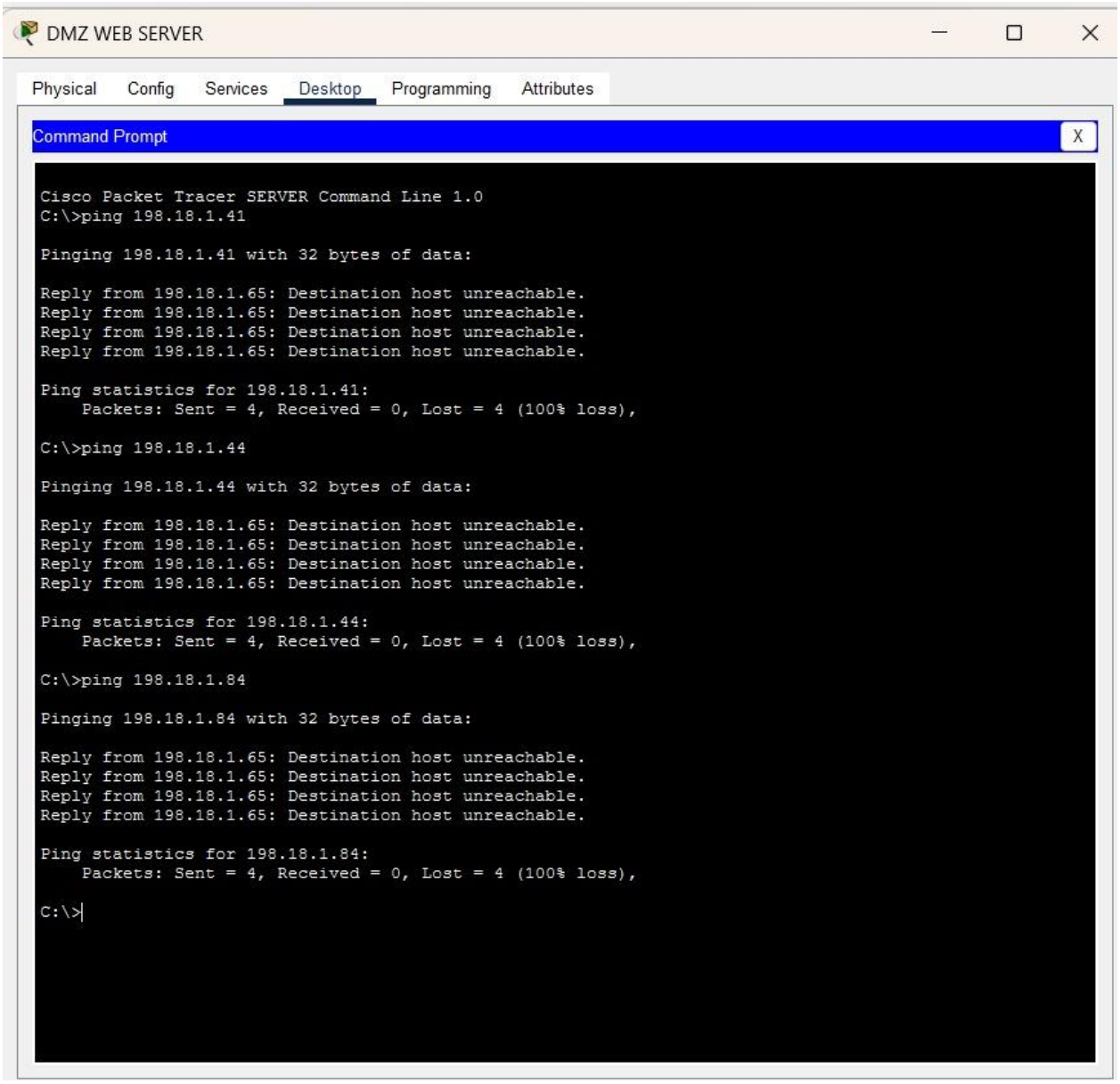


Figure 54 : DMZ Isolation from Sensitive Internal Hospital Systems

14. REMOTE AMBULANCE CONNECTIVITY AND ACCESS CONTROL

The remote ambulance uses the following network:

- Ambulance LAN : 10.100.0.0/24
- AMBULANCE-RTR : 10.100.0.1
- AMBULANCE-PC : 10.100.0.10
- GRE Tunnel : 10.200.0.0/30
- Ambulance Tunnel IP : 10.200.0.1
- Hospital Tunnel IP : 10.200.0.2

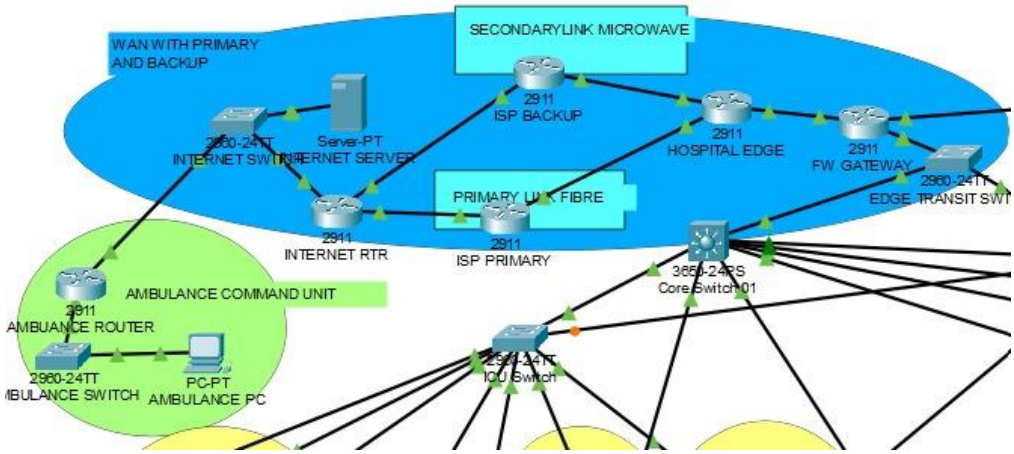


Figure 55 : Remote Ambulance Connectivity To The Hospital Network

14.1 Authorized Clinical Access

The ambulance is permitted to access only the hospital services required for patient information exchange and telemetry.

```
C:\>ping 198.18.1.41

Pinging 198.18.1.41 with 32 bytes of data:

Reply from 198.18.1.41: bytes=32 time=1ms TTL=125
Reply from 198.18.1.41: bytes=32 time<1ms TTL=125
Reply from 198.18.1.41: bytes=32 time=9ms TTL=125
Reply from 198.18.1.41: bytes=32 time=9ms TTL=125

Ping statistics for 198.18.1.41:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 9ms, Average = 4ms

C:\>ping 198.18.1.44

Pinging 198.18.1.44 with 32 bytes of data:

Reply from 198.18.1.44: bytes=32 time<1ms TTL=125
Reply from 198.18.1.44: bytes=32 time=10ms TTL=125
Reply from 198.18.1.44: bytes=32 time<1ms TTL=125
Reply from 198.18.1.44: bytes=32 time<1ms TTL=125

Ping statistics for 198.18.1.44:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 2ms
```

Figure 56 : Successful Ambulance Access to EMR/EHR and Telemetry Services

14.3 Restricted Ambulance Access

The ambulance is not permitted to access unrelated sensitive hospital systems .

```
C:\>ping 198.18.1.40

Pinging 198.18.1.40 with 32 bytes of data:

Reply from 10.200.0.2: Destination host unreachable.
Reply from 10.200.0.2: Destination host unreachable.
Reply from 10.200.0.2: Destination host unreachable.
Reply from 10.200.0.2: Destination host unreachable.

Ping statistics for 198.18.1.40:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 198.18.1.21

Pinging 198.18.1.21 with 32 bytes of data:

Reply from 10.200.0.2: Destination host unreachable.
Reply from 10.200.0.2: Destination host unreachable.
Reply from 10.200.0.2: Destination host unreachable.
Reply from 10.200.0.2: Destination host unreachable.

Ping statistics for 198.18.1.21:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 198.18.1.84

Pinging 198.18.1.84 with 32 bytes of data:

Reply from 10.200.0.2: Destination host unreachable.
Reply from 10.200.0.2: Destination host unreachable.
Reply from 10.200.0.2: Destination host unreachable.
Reply from 10.200.0.2: Destination host unreachable.

Ping statistics for 198.18.1.84:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Figure 57 : Restricted Ambulance Access to PACS, HOC and Management Networks

14.4 Access-Control Verification

```
Router>enable
Router#show access-lists AMBULANCE-VPN
Extended IP access list AMBULANCE-VPN
    permit ip 10.100.0.0 0.0.0.255 host 198.18.1.41 (7 match(es))
    permit ip 10.100.0.0 0.0.0.255 host 198.18.1.44 (4 match(es))
    deny ip 10.100.0.0 0.0.0.255 198.18.0.0 0.0.1.255 (12 match(es))
    permit ip any any (37 match(es))

Router#
```

Figure 58 : Ambulance Access-Control Policy on FW-GATEWAY

15. DHCP AND WIRELESS NETWORK IMPLEMENTATION

15.1 DHCP Implementation

DHCP was configured to automatically assign IPv4 addresses to client devices in the hospital VLANs. The configured DHCP pools cover the clinical wards, Administration, and HOC networks.

Core Switch 01

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch#show ip dhcp pool

Pool ICU_POOL :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 62
 Leased addresses               : 7
 Excluded addresses             : 11
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 198.18.0.1           198.18.0.1       - 198.18.0.62      7    / 11    / 62

Pool CARDIO_POOL :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 62
 Leased addresses               : 7
 Excluded addresses             : 11
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 198.18.0.65          198.18.0.65      - 198.18.0.126     7    / 11    / 62

Pool EMERG_POOL :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 62
 Leased addresses               : 7
 Excluded addresses             : 11
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 198.18.0.129         198.18.0.129     - 198.18.0.190     7    / 11    / 62

Pool OUTPATIENT_POOL :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 62
 Leased addresses               : 7
 Excluded addresses             : 11
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 198.18.0.193         198.18.0.193     - 198.18.0.254     7    / 11    / 62

 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 198.18.0.193         198.18.0.193     - 198.18.0.254     7    / 11    / 62

Pool ADMIN_POOL :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 14
 Leased addresses               : 3
 Excluded addresses             : 11
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 198.18.1.1           198.18.1.1       - 198.18.1.14      3    / 11    / 14

Pool HOC_POOL :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 14
 Leased addresses               : 3
 Excluded addresses             : 11
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 198.18.1.17          198.18.1.17      - 198.18.1.30      3    / 11    / 14
```

Figure 59 : DHCP Pools Configured for Hospital VLANs

15.2 DHCP Client Verification

```
Switch#show ip dhcp binding
IP address       Client-ID/              Lease expiration        Type
                 Hardware address
198.18.0.12      0001.9655.C732           --                     Automatic
198.18.0.10      00D0.D34C.03BA           --                     Automatic
198.18.0.11      00E0.8F0D.4827           --                     Automatic
198.18.0.13      00D0.D3EB.B0D2           --                     Automatic
198.18.0.14      000A.F379.95C6           --                     Automatic
198.18.0.15      0004.9A97.B2D0           --                     Automatic
198.18.0.16      0005.5E6D.9D5E           --                     Automatic
198.18.0.70      000C.CF11.3050           --                     Automatic
198.18.0.71      0001.4278.202A           --                     Automatic
198.18.0.72      0001.C730.84DB           --                     Automatic
198.18.0.74      0060.3E3C.7D69           --                     Automatic
198.18.0.73      000A.F334.B5AB           --                     Automatic
198.18.0.75      00D0.5883.7025           --                     Automatic
198.18.0.76      000C.8540.77B9           --                     Automatic
198.18.0.134     000D.BDDD.41D9           --                     Automatic
198.18.0.136     0040.0BC6.0DA1           --                     Automatic
198.18.0.137     00D0.BCEA.D9D8           --                     Automatic
198.18.0.135     0050.0F55.2256           --                     Automatic
198.18.0.138     0001.C7AC.EDDB           --                     Automatic
198.18.0.139     0005.5E02.CC1E           --                     Automatic
198.18.0.140     00D0.9708.5ACD           --                     Automatic
198.18.0.198     0090.2B50.B265           --                     Automatic
198.18.0.200     00D0.97D8.4828           --                     Automatic
198.18.0.201     0002.1686.699C           --                     Automatic
198.18.0.202     000A.F3D5.9540           --                     Automatic
198.18.0.199     00D0.BCC9.1279           --                     Automatic
198.18.0.203     0090.0CC5.1BD6           --                     Automatic
198.18.0.204     00E0.A306.0B89           --                     Automatic
198.18.1.5       00E0.F722.4D8E           --                     Automatic
198.18.1.6       0050.0F65.82B4           --                     Automatic
198.18.1.7       0060.7009.0D5A           --                     Automatic
198.18.1.23      000C.8544.4D25           --                     Automatic
198.18.1.22      0007.ECAD.A0D7           --                     Automatic
198.18.1.21      000A.F34A.C7AB           --                     Automatic
Switch#
```

Figure 60 : Active DHCP Address Bindings

```
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.....: FE80::201:96FF:FE55:C732
   IPv6 Address.....: 2001:DB8:ABCD:10:201:96FF:FE55:C732
   IPv4 Address.....: 198.18.0.12
   Subnet Mask.....: 255.255.255.192
   Default Gateway.....: FE80::206:2AFF:FED5:E401
                         198.18.0.1

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.....: ::
   IPv6 Address.....: ::
   IPv4 Address.....: 0.0.0.0
   Subnet Mask.....: 0.0.0.0
   Default Gateway.....: ::
                         0.0.0.0
```

Figure 61 : ICU Workstation Receiving an IPv4 Address Through DHCP

15.3 Wireless Network Implementation

Wireless connectivity was provided for clinical staff using access points connected to the corresponding ward networks.

The wireless design supports staff tablets and laptops while maintaining separation between the different clinical areas.

Table 18 : Wireless Network Implementation

Ward	SSID
ICU / Surgical	ICU-CLINICAL
Cardiology	CARDIO-CLINICAL
Emergency	EMERG-CLINICAL
Outpatient	OUTPATIENT-CLINICAL

15.4 Wireless Client Verification

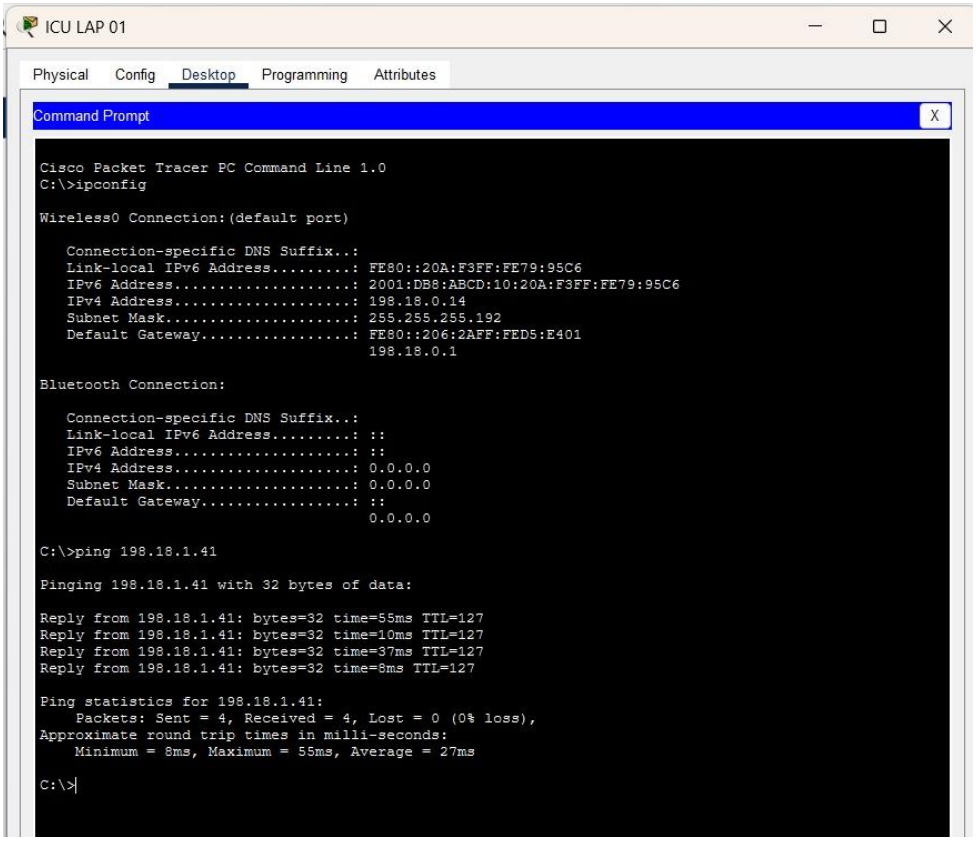


Figure 62 : Clinical Wireless Client Connected to the Ward Access Point and Successful Wireless Access to the EMR Server

16. NETWORK REDUNDANCY AND HIGH AVAILABILITY

16.1 Redundant Core and Default Gateway

```
Switch#show standby brief
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State    Active          Standby         Virtual IP
Vl10        10   110 P Active   local           198.18.0.3      198.18.0.1
Vl20        20   110 P Active   local           198.18.0.67     198.18.0.65
Vl30        30   110 P Active   local           198.18.0.131    198.18.0.129
Vl40        40   110 P Active   local           198.18.0.195    198.18.0.193
Vl50        50   110 P Active   local           198.18.1.3      198.18.1.1
Vl60        60   110 P Active   local           198.18.1.19     198.18.1.17
Vl70        70   110 P Active   local           198.18.1.35     198.18.1.33
Vl90        90   110 P Active   local           198.18.1.83     198.18.1.81
Vl999       99   110 P Active   local           10.255.0.3      10.255.0.1
```

Figure 63 : HSRP-Based Gateway Redundancy on the Hospital Core

16.2 Primary and Secondary WAN Redundancy

The hospital uses:

- Primary → Dedicated Fibre
- Secondary → Microwave

OSPF is configured to prefer the primary route while retaining the secondary connection as an alternative.

The preferred costs are:

- Primary Fibre → Cost 10
- Backup Microwave → Cost 100

This allows traffic to automatically move to the secondary path if the primary link becomes unavailable.

16.3 WAN Failover Verification

```
Router>
Router>enable
Router#show ip route 192.0.2.128
Routing entry for 192.0.2.128/25
Known via "ospf 1", distance 110, metric 12, type intra area
  Last update from 203.0.113.2 on GigabitEthernet0/1, 01:40:45 ago
  Routing Descriptor Blocks:
  * 203.0.113.2, from 192.0.2.129, 01:40:45 ago, via GigabitEthernet0/1
      Route metric is 12, traffic share count is 1
```

Figure 64 : External Route Using the Primary Fibre Connection

```
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gi0/1
Router(config-if)#shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

08:29:11: %OSPF-5-ADJCHG: Process 1, Nbr 203.0.113.2 on GigabitEthernet0/1 from FULL to DOWN,
Neighbor Down: Interface down or detached
end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip route 192.0.2.128
Routing entry for 192.0.2.128/25
Known via "ospf 1", distance 110, metric 102, type intra area
  Last update from 198.51.100.2 on GigabitEthernet0/2, 00:00:09 ago
  Routing Descriptor Blocks:
  * 198.51.100.2, from 192.0.2.129, 00:00:09 ago, via GigabitEthernet0/2
      Route metric is 102, traffic share count is 1
```

Figure 65 : Shutdown The Primary Link

```
C:\>ping 192.0.2.130

Pinging 192.0.2.130 with 32 bytes of data:

Reply from 192.0.2.130: bytes=32 time<1ms TTL=123
Reply from 192.0.2.130: bytes=32 time=14ms TTL=123
Reply from 192.0.2.130: bytes=32 time<1ms TTL=123
Reply from 192.0.2.130: bytes=32 time<1ms TTL=123

Ping statistics for 192.0.2.130:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 14ms, Average = 3ms

C:\>
```

Figure 66: Automatic Failover to the Secondary Microwave Connection and testing from the ICU PC

16.4 Layer 2 Redundancy

Spanning Tree Protocol is used where redundant switching paths exist to prevent Layer 2 loops.

```
Switch#show spanning-tree vlan 90
VLAN0090
  Spanning tree enabled protocol ieee
  Root ID     Priority    24666
              Address     0006.2AD5.E4E1
              This bridge is the root
              Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID   Priority    24666  (priority 24576 sys-id-ext 90)
              Address     0006.2AD5.E4E1
              Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Gi1/0/1          Desg FWD 19        128.1    P2p
Gi1/0/2          Desg FWD 4         128.2    P2p
Gi1/0/3          Desg FWD 19        128.3    P2p
Gi1/0/4          Desg FWD 19        128.4    P2p
Gi1/0/5          Desg FWD 19        128.5    P2p
Gi1/0/6          Desg FWD 19        128.6    P2p
Gi1/0/7          Desg FWD 19        128.7    P2p
```

Figure 67 : Spanning Tree Operation on the Hospital Switching Infrastructure

17. ROUTING TABLES FOR ALL ROUTERS

```
Switch#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.255.0.4 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.255.0.0/29 is directly connected, Vlan999
L       10.255.0.2/32 is directly connected, Vlan999
     198.18.0.0/24 is variably subnetted, 8 subnets, 2 masks
C       198.18.0.0/26 is directly connected, Vlan10
L       198.18.0.2/32 is directly connected, Vlan10
C       198.18.0.64/26 is directly connected, Vlan20
L       198.18.0.66/32 is directly connected, Vlan20
C       198.18.0.128/26 is directly connected, Vlan30
L       198.18.0.130/32 is directly connected, Vlan30
C       198.18.0.192/26 is directly connected, Vlan40
L       198.18.0.194/32 is directly connected, Vlan40
     198.18.1.0/24 is variably subnetted, 8 subnets, 3 masks
C       198.18.1.0/28 is directly connected, Vlan50
L       198.18.1.2/32 is directly connected, Vlan50
C       198.18.1.16/28 is directly connected, Vlan60
L       198.18.1.18/32 is directly connected, Vlan60
C       198.18.1.32/27 is directly connected, Vlan70
L       198.18.1.34/32 is directly connected, Vlan70
C       198.18.1.80/28 is directly connected, Vlan90
L       198.18.1.82/32 is directly connected, Vlan90
S*   0.0.0.0/0 [1/0] via 10.255.0.4
```

Figure 68 : Routing Table of CORE-SW1

```
Switch>
Switch>
Switch>enable
Switch#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.255.0.4 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.255.0.0/29 is directly connected, Vlan999
L       10.255.0.3/32 is directly connected, Vlan999
     198.18.0.0/24 is variably subnetted, 8 subnets, 2 masks
C       198.18.0.0/26 is directly connected, Vlan10
L       198.18.0.3/32 is directly connected, Vlan10
C       198.18.0.64/26 is directly connected, Vlan20
L       198.18.0.67/32 is directly connected, Vlan20
C       198.18.0.128/26 is directly connected, Vlan30
L       198.18.0.131/32 is directly connected, Vlan30
C       198.18.0.192/26 is directly connected, Vlan40
L       198.18.0.195/32 is directly connected, Vlan40
     198.18.1.0/24 is variably subnetted, 8 subnets, 3 masks
C       198.18.1.0/28 is directly connected, Vlan50
L       198.18.1.3/32 is directly connected, Vlan50
C       198.18.1.16/28 is directly connected, Vlan60
L       198.18.1.19/32 is directly connected, Vlan60
C       198.18.1.32/27 is directly connected, Vlan70
L       198.18.1.35/32 is directly connected, Vlan70
C       198.18.1.80/28 is directly connected, Vlan90
L       198.18.1.83/32 is directly connected, Vlan90
S*   0.0.0.0/0 [1/0] via 10.255.0.4
```

Figure 70 : Routing Table of CORE-SW2

```
Router>
Router>enable
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.255.0.10 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 7 subnets, 4 masks
S       10.100.0.0/24 [1/0] via 10.200.0.1
C       10.200.0.0/30 is directly connected, Tunnel0
L       10.200.0.2/32 is directly connected, Tunnel0
C       10.255.0.0/29 is directly connected, GigabitEthernet0/0
L       10.255.0.4/32 is directly connected, GigabitEthernet0/0
C       10.255.0.8/30 is directly connected, GigabitEthernet0/1
L       10.255.0.9/32 is directly connected, GigabitEthernet0/1
S    198.18.0.0/23 [1/0] via 10.255.0.1
     198.18.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       198.18.1.64/28 is directly connected, GigabitEthernet0/2
L       198.18.1.65/32 is directly connected, GigabitEthernet0/2
S*   0.0.0.0/0 [1/0] via 10.255.0.10

Router#
Router#
Router#
Router#
```

Figure 71 : Routing Table of FW-GATEWAY

```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.255.0.8/30 is directly connected, GigabitEthernet0/0
L       10.255.0.10/32 is directly connected, GigabitEthernet0/0
     192.0.2.0/24 is variably subnetted, 3 subnets, 2 masks
O       192.0.2.0/30 [110/11] via 203.0.113.2, 00:05:17, GigabitEthernet0/1
O       192.0.2.4/30 [110/101] via 198.51.100.2, 01:49:42, GigabitEthernet0/2
O       192.0.2.128/25 [110/12] via 203.0.113.2, 00:05:17, GigabitEthernet0/1
S    198.18.0.0/23 [1/0] via 10.255.0.9
     198.51.100.0/24 is variably subnetted, 2 subnets, 2 masks
C       198.51.100.0/30 is directly connected, GigabitEthernet0/2
L       198.51.100.1/32 is directly connected, GigabitEthernet0/2
     203.0.113.0/24 is variably subnetted, 2 subnets, 2 masks
C       203.0.113.0/30 is directly connected, GigabitEthernet0/1
L       203.0.113.1/32 is directly connected, GigabitEthernet0/1
```

Figure 72 : Routing Table of HOSPITAL-EDGE

```
Router>
Router>enable
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/30 is subnetted, 1 subnets
O       10.255.0.8/30 [110/2] via 203.0.113.1, 00:06:11, GigabitEthernet0/0
     192.0.2.0/24 is variably subnetted, 4 subnets, 3 masks
C       192.0.2.0/30 is directly connected, GigabitEthernet0/1
L       192.0.2.1/32 is directly connected, GigabitEthernet0/1
O       192.0.2.4/30 [110/101] via 192.0.2.2, 01:50:36, GigabitEthernet0/1
O       192.0.2.128/25 [110/2] via 192.0.2.2, 01:50:36, GigabitEthernet0/1
O E2 198.18.0.0/23 [110/20] via 203.0.113.1, 00:06:11, GigabitEthernet0/0
     198.51.100.0/30 is subnetted, 1 subnets
O       198.51.100.0/30 [110/101] via 203.0.113.1, 00:06:11, GigabitEthernet0/0
     203.0.113.0/24 is variably subnetted, 2 subnets, 2 masks
C       203.0.113.0/30 is directly connected, GigabitEthernet0/0
L       203.0.113.2/32 is directly connected, GigabitEthernet0/0
```

Figure 73 : Routing Table of ISP-PRIMARY

```
Router>enable
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/30 is subnetted, 1 subnets
O       10.255.0.8/30 [110/2] via 198.51.100.1, 01:52:11, GigabitEthernet0/0
     192.0.2.0/24 is variably subnetted, 4 subnets, 3 masks
O       192.0.2.0/30 [110/11] via 192.0.2.6, 01:52:11, GigabitEthernet0/1
C       192.0.2.4/30 is directly connected, GigabitEthernet0/1
L       192.0.2.5/32 is directly connected, GigabitEthernet0/1
O       192.0.2.128/25 [110/2] via 192.0.2.6, 01:52:11, GigabitEthernet0/1
O E2 198.18.0.0/23 [110/20] via 198.51.100.1, 01:52:11, GigabitEthernet0/0
     198.51.100.0/24 is variably subnetted, 2 subnets, 2 masks
C       198.51.100.0/30 is directly connected, GigabitEthernet0/0
L       198.51.100.2/32 is directly connected, GigabitEthernet0/0
     203.0.113.0/30 is subnetted, 1 subnets
O       203.0.113.0/30 [110/11] via 198.51.100.1, 00:07:46, GigabitEthernet0/0
```

Figure 74: Routing Table of ISP-BACKUP

```
Router>enable
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/30 is subnetted, 1 subnets
O       10.255.0.8/30 [110/12] via 192.0.2.1, 00:08:34, GigabitEthernet0/0
     192.0.2.0/24 is variably subnetted, 6 subnets, 3 masks
C       192.0.2.0/30 is directly connected, GigabitEthernet0/0
L       192.0.2.2/32 is directly connected, GigabitEthernet0/0
C       192.0.2.4/30 is directly connected, GigabitEthernet0/1
L       192.0.2.6/32 is directly connected, GigabitEthernet0/1
C       192.0.2.128/25 is directly connected, GigabitEthernet0/2
L       192.0.2.129/32 is directly connected, GigabitEthernet0/2
O E2 198.18.0.0/23 [110/20] via 192.0.2.1, 00:08:34, GigabitEthernet0/0
     198.51.100.0/30 is subnetted, 1 subnets
O       198.51.100.0/30 [110/101] via 192.0.2.5, 01:52:59, GigabitEthernet0/1
     203.0.113.0/30 is subnetted, 1 subnets
O       203.0.113.0/30 [110/11] via 192.0.2.1, 00:09:14, GigabitEthernet0/0
```

Figure 75 : Routing Table Of INTERNET-RTR

```
AMBULANCE-RTR>
AMBULANCE-RTR>enable
AMBULANCE-RTR#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 192.0.2.129 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 4 subnets, 3 masks
C       10.100.0.0/24 is directly connected, GigabitEthernet0/1
L       10.100.0.1/32 is directly connected, GigabitEthernet0/1
C       10.200.0.0/30 is directly connected, Tunnel0
L       10.200.0.1/32 is directly connected, Tunnel0
     192.0.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.0.2.128/25 is directly connected, GigabitEthernet0/0
L       192.0.2.131/32 is directly connected, GigabitEthernet0/0
S    198.18.0.0/23 [1/0] via 10.200.0.2
S*   0.0.0.0/0 [1/0] via 192.0.2.129
```

Figure 76 : Routing Table of the Remote Ambulance Router

18. TESTING AND VERIFICATION RESULTS

```
C:\>ping 198.18.1.41

Pinging 198.18.1.41 with 32 bytes of data:

Reply from 198.18.1.41: bytes=32 time=1ms TTL=127
Reply from 198.18.1.41: bytes=32 time<1ms TTL=127
Reply from 198.18.1.41: bytes=32 time<1ms TTL=127
Reply from 198.18.1.41: bytes=32 time<1ms TTL=127

Ping statistics for 198.18.1.41:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Figure 77: Successful IPv4 Connectivity Between ICU and EMR Server

The successful ICMP replies confirm that the ICU network can communicate with the centralized EMR/EHR server. This verifies correct VLAN assignment, inter-VLAN routing, gateway configuration, and Data Centre connectivity.

```
C:\>ping 2001:DB8:ABCD:70::41

Pinging 2001:DB8:ABCD:70::41 with 32 bytes of data:

Reply from 2001:DB8:ABCD:70::41: bytes=32 time=1ms TTL=127
Reply from 2001:DB8:ABCD:70::41: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:ABCD:70::41: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:ABCD:70::41: bytes=32 time<1ms TTL=127

Ping statistics for 2001:DB8:ABCD:70::41:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Figure 78 : Successful IPv6 Connectivity Between ICU and EMR Server

The successful IPv6 ping verifies that the hospital network operates using dual-stack IPv4/IPv6 addressing and that IPv6 routing between the clinical VLAN and Server VLAN is functioning correctly.

```
Switch#show ip dhcp binding
IP address       Client-ID/              Lease expiration        Type
                 Hardware address
198.18.0.12      0001.9655.C732           --                     Automatic
198.18.0.10      00D0.D34C.03BA           --                     Automatic
198.18.0.11      00E0.8F0D.4827           --                     Automatic
198.18.0.13      00D0.D3EB.B0D2           --                     Automatic
198.18.0.14      000A.F379.95C6           --                     Automatic
198.18.0.15      0004.9A97.B2D0           --                     Automatic
198.18.0.16      0005.5E6D.9D5E           --                     Automatic
198.18.0.70      000C.CF11.3050           --                     Automatic
198.18.0.71      0001.4278.202A           --                     Automatic
198.18.0.72      0001.C730.84DB           --                     Automatic
198.18.0.74      0060.3E3C.7D69           --                     Automatic
198.18.0.73      000A.F334.B5AB           --                     Automatic
198.18.0.75      00D0.5883.7025           --                     Automatic
198.18.0.76      000C.8540.77B9           --                     Automatic
198.18.0.134     000D.BDDD.41D9           --                     Automatic
198.18.0.136     0040.0BC6.0DA1           --                     Automatic
198.18.0.137     00D0.BCEA.D9D8           --                     Automatic
198.18.0.135     0050.0F55.2256           --                     Automatic
198.18.0.138     0001.C7AC.EDDB           --                     Automatic
198.18.0.139     0005.5E02.CC1E           --                     Automatic
198.18.0.140     00D0.9708.5ACD           --                     Automatic
198.18.0.198     0090.2B50.B265           --                     Automatic
198.18.0.200     00D0.97D8.4828           --                     Automatic
198.18.0.201     0002.1686.699C           --                     Automatic
198.18.0.202     000A.F3D5.9540           --                     Automatic
198.18.0.199     00D0.BCC9.1279           --                     Automatic
198.18.0.203     0090.0CC5.1BD6           --                     Automatic
198.18.0.204     00E0.A306.0B89           --                     Automatic
198.18.1.5       00E0.F722.4D8E           --                     Automatic
198.18.1.6       0050.0F65.82B4           --                     Automatic
198.18.1.7       0060.7009.0D5A           --                     Automatic
198.18.1.23      000C.8544.4D25           --                     Automatic
198.18.1.22      0007.ECAD.A0D7           --                     Automatic
198.18.1.21      000A.F34A.C7AB           --                     Automatic
Switch#
```

Figure 79 : DHCP Address Assignment to Hospital Clients

The DHCP binding table and client configuration confirm that hospital endpoints automatically receive valid IPv4 addresses, subnet masks, and default gateways from the correct departmental address pools.

```
Switch#show standby brief
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State    Active          Standby         Virtual IP
Vl10        10   110 P Active   local           198.18.0.3      198.18.0.1
Vl20        20   110 P Active   local           198.18.0.67     198.18.0.65
Vl30        30   110 P Active   local           198.18.0.131    198.18.0.129
Vl40        40   110 P Active   local           198.18.0.195    198.18.0.193
Vl50        50   110 P Active   local           198.18.1.3      198.18.1.1
Vl60        60   110 P Active   local           198.18.1.19     198.18.1.17
Vl70        70   110 P Active   local           198.18.1.35     198.18.1.33
Vl90        90   110 P Active   local           198.18.1.83     198.18.1.81
Vl1999      99   110 P Active   local           10.255.0.3      10.255.0.1
Switch#
Switch#
```

Figure 80: HSRP Gateway Redundancy on the Core Network

The HSRP output confirms that redundant virtual gateways are configured for the hospital VLANs. CORE-SW1 operates as the preferred active gateway while CORE-SW2 provides standby gateway functionality

```
C:\>ping 192.0.2.130

Pinging 192.0.2.130 with 32 bytes of data:

Reply from 10.255.0.4: Destination host unreachable.
Reply from 10.255.0.4: Destination host unreachable.
Reply from 10.255.0.4: Destination host unreachable.
Reply from 10.255.0.4: Destination host unreachable.

Ping statistics for 192.0.2.130:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Figure 81 : Administration Internet Access Successfully Restricted

The failed ping confirms that Administration users are prevented from obtaining direct outbound Internet access, satisfying the security requirement while retaining access to approved internal services.

```
C:\>ping 198.18.1.84

Pinging 198.18.1.84 with 32 bytes of data:

Reply from 198.18.1.84: bytes=32 time<1ms TTL=254
Reply from 198.18.1.84: bytes=32 time<1ms TTL=254
Reply from 198.18.1.84: bytes=32 time<1ms TTL=254
Reply from 198.18.1.84: bytes=32 time<1ms TTL=254

Ping statistics for 198.18.1.84:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Figure 82 : Authorized HOC Access to the Management Network

The successful communication demonstrates that authorized HOC workstations can reach network-management interfaces in VLAN 90. This confirms that management access is available to the appropriate administrative users.

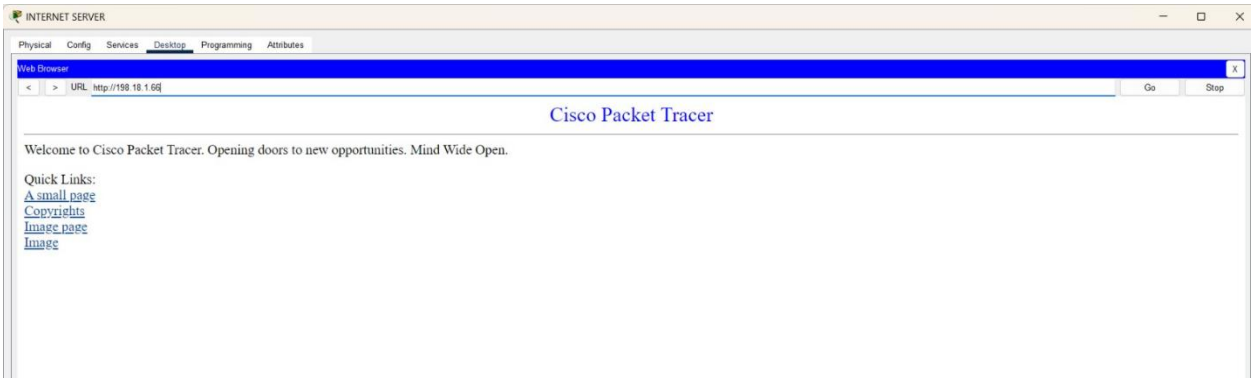


Figure 83: Successful External Access to the DMZ Web Server

The web page successfully loaded from the DMZ server, confirming that permitted public-facing services can be accessed from the external network.

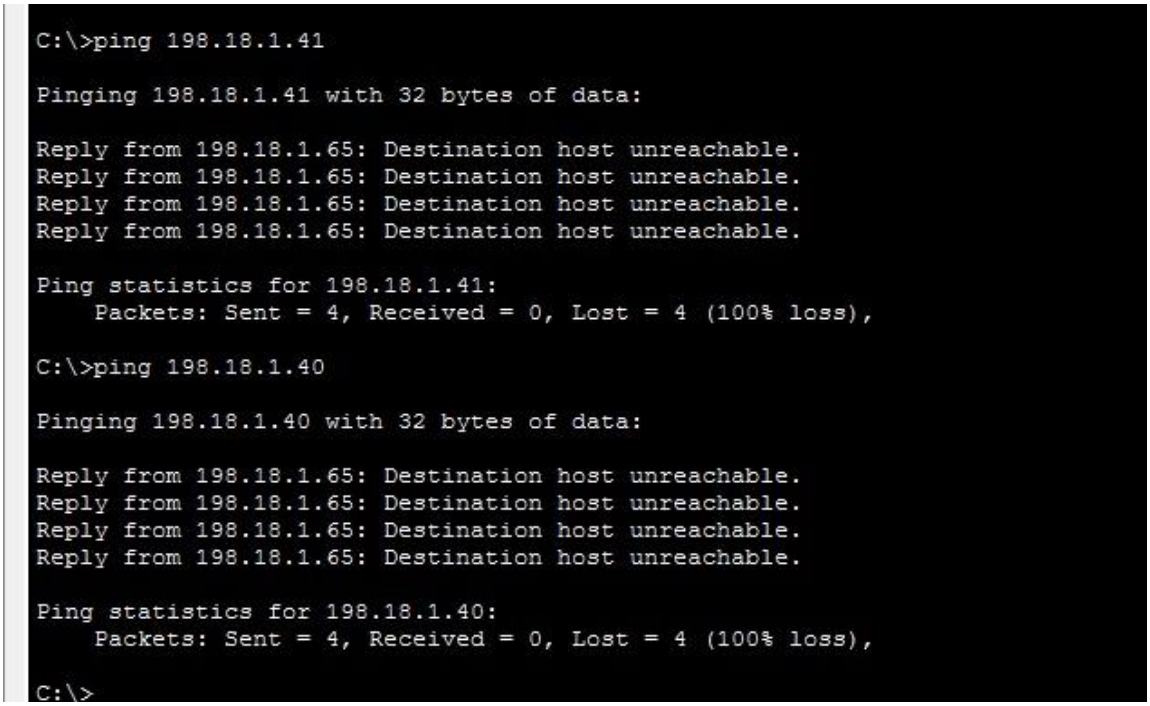


Figure 84 : DMZ Isolation from EMR and PACS Servers

The failed connectivity tests confirm that a host located in the DMZ cannot initiate communication toward sensitive internal hospital servers. This verifies correct DMZ isolation.

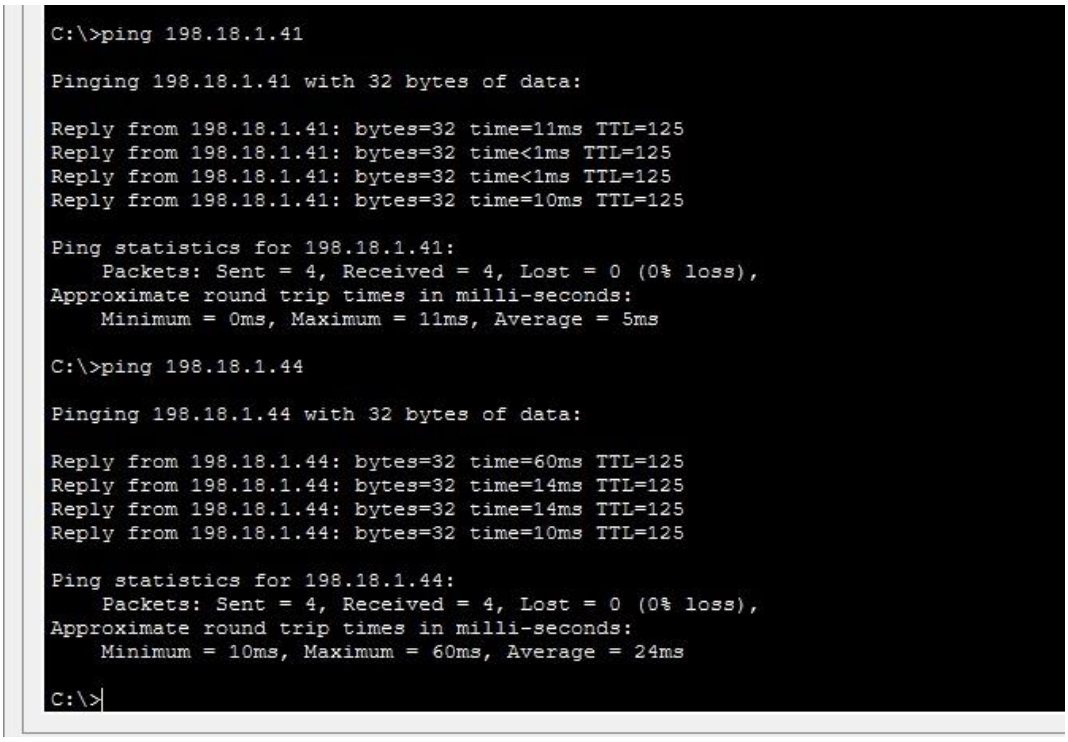


Figure 85 : Successful Ambulance Access to EMR and Telemetry Services

Successful communication confirms that the remote ambulance network can access the approved clinical services required for patient information and telemetry exchange.

```
C:\>ping 198.18.1.40

Pinging 198.18.1.40 with 32 bytes of data:

Reply from 10.200.0.2: Destination host unreachable.
Reply from 10.200.0.2: Destination host unreachable.
Reply from 10.200.0.2: Destination host unreachable.
Reply from 10.200.0.2: Destination host unreachable.

Ping statistics for 198.18.1.40:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Figure 86 : Restricted Ambulance Access to PACS

The failed ping confirms that the remote ambulance network is prevented from accessing unrelated sensitive hospital systems, demonstrating correct access-control enforcement.

```
AMBULANCE-RTR>enable
AMBULANCE-RTR#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.0.2.131     YES manual up                    up
GigabitEthernet0/1     10.100.0.1      YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Tunnel0                10.200.0.1      YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
AMBULANCE-RTR#
```

Figure 87 : Operational GRE Tunnel Between Ambulance and Hospital

Tunnel0 appearing in the up/up state confirms that the logical tunnel between the ambulance router and hospital firewall gateway is operational.

```
Router#show ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
icmp 10.255.0.9:17     198.18.0.30:17     192.0.2.130:17     192.0.2.130:17
icmp 10.255.0.9:18     198.18.0.30:18     192.0.2.130:18     192.0.2.130:18
icmp 10.255.0.9:19     198.18.0.30:19     192.0.2.130:19     192.0.2.130:19
icmp 10.255.0.9:20     198.18.0.30:20     192.0.2.130:20     192.0.2.130:20
```

```
Router#show ip nat statistics
Total translations: 0 (0 static, 0 dynamic, 0 extended)
Outside Interfaces: GigabitEthernet0/1
Inside Interfaces: GigabitEthernet0/0
Hits: 13  Misses: 15
Expired translations: 15
Dynamic mappings:
Router#show access-lists 1
Standard IP access list 1
    permit 198.18.0.0 0.0.1.255 (30 match(es))
```

Figure 88: NAT/PAT Translation of Hospital Traffic

The translation table shows the internal source address being translated before communication with the external network. This confirms that NAT/PAT is functioning on FW-GATEWAY.

```
Router>
Router>enable
Router#show ip route 192.0.2.128
Routing entry for 192.0.2.128/25
Known via "ospf 1", distance 110, metric 12, type intra area
  Last update from 203.0.113.2 on GigabitEthernet0/1, 00:47:42 ago
  Routing Descriptor Blocks:
  * 203.0.113.2, from 192.0.2.129, 00:47:42 ago, via GigabitEthernet0/1
      Route metric is 12, traffic share count is 1

Router#
```

Figure 89 : Primary Fibre Route Selected During Normal Operation

The routing table confirms that the external network is normally reached through the preferred primary fibre path because of its lower routing cost.

```
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gi0/1
Router(config-if)#shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

05:39:17: %OSPF-5-ADJCHG: Process 1, Nbr 203.0.113.2 on GigabitEthernet0/1 from FULL to DOWN,
Neighbor Down: Interface down or detached

Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

```
Router#show ip route 192.0.2.128
Routing entry for 192.0.2.128/25
Known via "ospf 1", distance 110, metric 102, type intra area
  Last update from 198.51.100.2 on GigabitEthernet0/2, 00:00:26 ago
  Routing Descriptor Blocks:
  * 198.51.100.2, from 192.0.2.129, 00:00:26 ago, via GigabitEthernet0/2
      Route metric is 102, traffic share count is 1

Router#
```

```
Router#ping 192.0.2.130

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.0.2.130, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

Router#
```

Figure 90: Automatic Failover to the Secondary Microwave Link

After the primary WAN interface was disabled, the routing table selected the secondary path and external connectivity remained available. This verifies WAN redundancy and automatic failover**.**

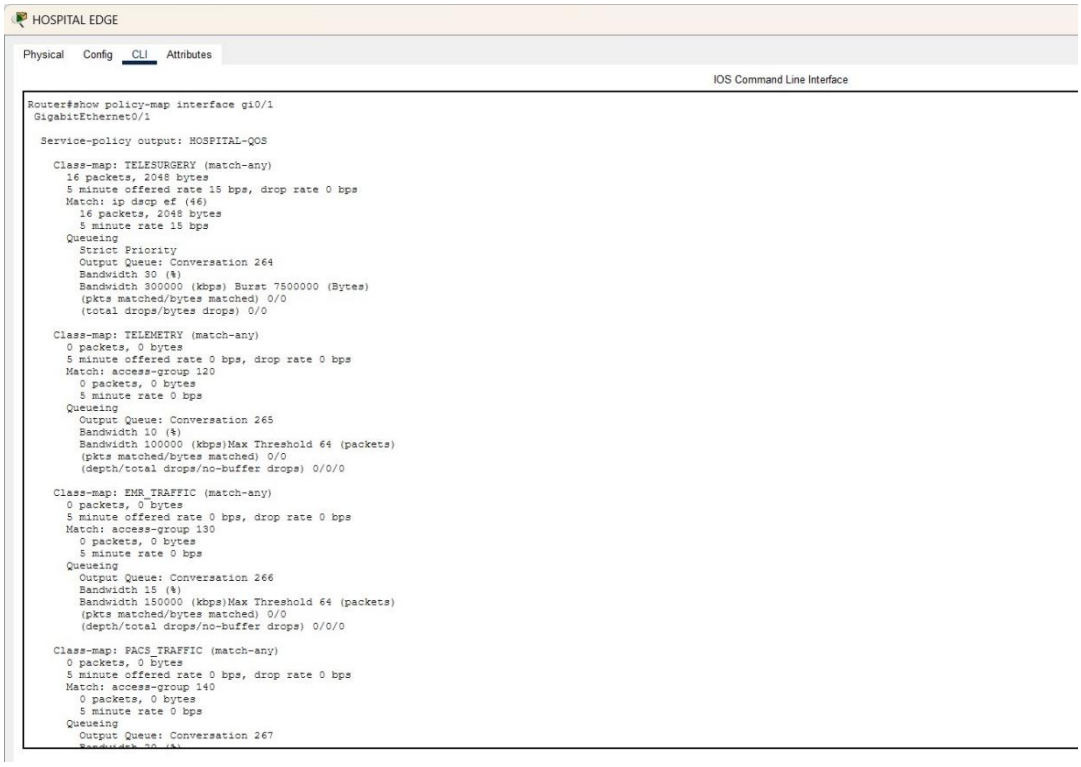


Figure 91: Hospital QoS Policy Applied to the Primary WAN Interface

The output confirms that the HOSPITAL-QOS policy is attached to the WAN interface and contains separate traffic classes for telesurgery, telemetry, EMR, and PACS. This verifies the configured traffic-prioritization policy.

```
Router#show access-lists DMZ-TO-INTERNAL
Extended IP access list DMZ-TO-INTERNAL
    deny ip 198.18.1.64 0.0.0.15 198.18.0.0 0.0.1.255 (28 match(es))
    permit ip any any (9 match(es))

Router#show access-lists
Standard IP access list 1
    10 permit 198.18.0.0 0.0.1.255 (94 match(es))
Extended IP access list OUTBOUND-POLICY
    10 deny ip 198.18.1.0 0.0.0.15 any (12 match(es))
    20 permit ip any any (6707 match(es))
Extended IP access list AMBULANCE-VPN
    10 permit ip 10.100.0.0 0.0.0.255 host 198.18.1.41 (15 match(es))
    20 permit ip 10.100.0.0 0.0.0.255 host 198.18.1.44 (12 match(es))
    30 deny ip 10.100.0.0 0.0.0.255 198.18.0.0 0.0.1.255 (20 match(es))
    40 permit ip any any (45 match(es))
Extended IP access list DMZ-TO-INTERNAL
    10 deny ip 198.18.1.64 0.0.0.15 198.18.0.0 0.0.1.255 (28 match(es))
    20 permit ip any any (9 match(es))
Extended IP access list TO-DMZ
    10 permit tcp any host 198.18.1.66 eq www (16 match(es))
    20 permit tcp any host 198.18.1.66 eq 443
    30 permit icmp host 192.0.2.130 host 198.18.1.66
    40 permit ip 198.18.0.0 0.0.1.255 198.18.1.64 0.0.0.15
    50 deny ip any 198.18.1.64 0.0.0.15
    60 permit ip any any
Extended IP access list 111
    10 permit ip host 198.18.0.30 any
```

Figure 92: Firewall ACL Enforcement on FW-GATEWAY

The ACL output confirms that permit and deny rules are configured to control communication between trusted, DMZ, external, and remote networks. Packet counters provide evidence that the rules have processed traffic.

19. RESULTS, DISCUSSION AND LIMITATIONS

19.1 Results Discussion

The implemented Cisco Packet Tracer network successfully demonstrated the main functional requirements of the hospital design. The four clinical wards were separated using VLANs and were able to access centralized medical services such as EMR/EHR, PACS, and Telemetry through the multilayer core network.

Dual-stack operation was also verified, with successful IPv4 and IPv6 communication between clinical workstations and hospital servers. This confirms that the addressing and inter-VLAN routing design operates correctly for both protocol families.

The security policies behaved according to the intended access requirements. Administration users were able to access internal Billing and Inventory services while direct Internet access was

blocked. HOC workstations retained access to the management VLAN, whereas unauthorized clinical and administrative devices were restricted. The DMZ web server remained externally accessible while being isolated from sensitive internal servers. Similarly, the remote ambulance was permitted to access EMR/EHR and Telemetry services while access to PACS, HOC, and management resources was restricted.

The WAN redundancy test showed that the hospital could continue reaching the simulated external network after the primary fibre link was disabled. OSPF selected the secondary microwave route, demonstrating the intended backup-path operation.

NAT/PAT was also verified on FW-GATEWAY, where internal hospital addresses were translated before external communication. The QoS policy was successfully created and attached to the hospital WAN interfaces, with separate traffic classes for telesurgery, telemetry, EMR, and PACS.

19.2 Design Strengths

The main strengths of the proposed design are its segmentation, redundancy, and controlled access. Departmental VLANs reduce unnecessary broadcast traffic and provide clear security boundaries. The redundant core infrastructure and HSRP improve gateway availability, while the use of separate fibre and microwave WAN technologies reduces dependence on a single external path.

The centralized Data Centre provides common access to important hospital services, while the DMZ prevents public-facing services from being placed directly within sensitive internal networks. The combination of ACLs, management restrictions, NAT/PAT, and remote-access controls provides a layered security approach suitable for the simulated environment.

19.3 Recommended Real-World Improvements

For an actual hospital deployment, the simulated design should be extended with enterprise-grade firewalls, IDS/IPS, secure VPN technologies such as IPsec, WPA3-Enterprise with 802.1X authentication, centralized AAA, monitoring and logging, backup power systems, and redundant physical cabling.

Before supporting real robotic telesurgery, the network should also be tested using the actual medical platform to verify end-to-end latency, jitter, bandwidth, recovery time, and fail-safe behavior .

20. CONCLUSION

The proposed hospital network was successfully designed and implemented in Cisco Packet Tracer to support the communication requirements of the four clinical wards, Administration network, Hospital Operations Centre, centralized Data Centre, DMZ, external WAN, and remote ambulance network.

The final design provides separate VLANs for the major hospital departments, dual-stack IPv4/IPv6 connectivity, inter-VLAN routing, centralized medical servers, DHCP, wireless access, NAT/PAT, firewall-based access control, and remote ambulance connectivity. Redundant core switches with HSRP improve gateway availability, while OSPF and the combination of primary fibre and secondary microwave links provide external-path redundancy.

Critical healthcare services were also considered in the design. The network includes dedicated support for PACS/DICOM imaging, EMR/EHR access, patient telemetry, and robotic telesurgery traffic. QoS policies were configured to prioritize telesurgery, telemetry, EMR, and PACS traffic according to their operational importance.

Security was strengthened through network segmentation, restricted Administration Internet access, protected management access for HOC users, DMZ isolation, ambulance access-control policies, and firewall ACLs. Testing confirmed that authorized communication paths remained available while prohibited traffic was successfully blocked.

The Packet Tracer implementation demonstrates that the proposed architecture can provide a scalable, redundant, and secure logical network for a modern hospital environment. However, a real deployment would require additional validation using production hardware, enterprise-grade firewalls, encrypted VPN technologies, wireless site surveys, and application-level testing for latency-sensitive medical services.

Overall, the design meets the main assignment requirements while maintaining a balance between performance, scalability, security, redundancy, and cost efficiency.